SSLC
BIOLOGY
D+ MODULE- 2026
DDE KOLLAM

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JIMSY K DANIEL

CHAPTER-1

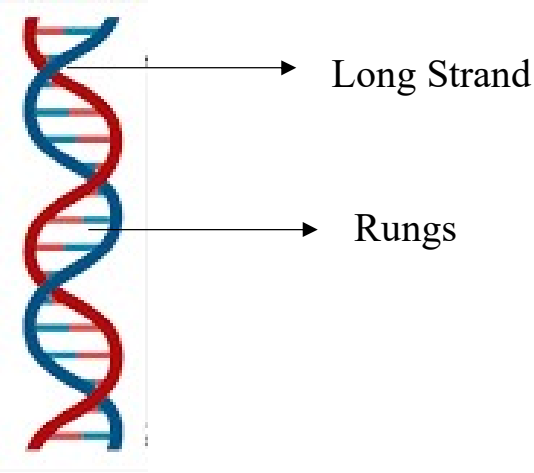
GENETICS OF LIFE

Location of DNA in cell

Cell → Nucleus → Chromosome → DNA.

STRUCTURE OF DNA

- In 1953 James Watson along with Francis Crick had presented the double helical model of DNA.
- The famous X-ray diffraction image taken by Rosalind Franklin known as 'photo 51' helped them to present this structure.
- The long strands on either side of DNA molecule is made up of sugar molecule and phosphate molecule.
- The rungs are made up of nitrogen base pairs.



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NITROGEN BASES IN DNA MOLECULE

- Adenine
- Thymine
- Guanine
- Cytosine

NITROGEN BASES AND PAIRING.

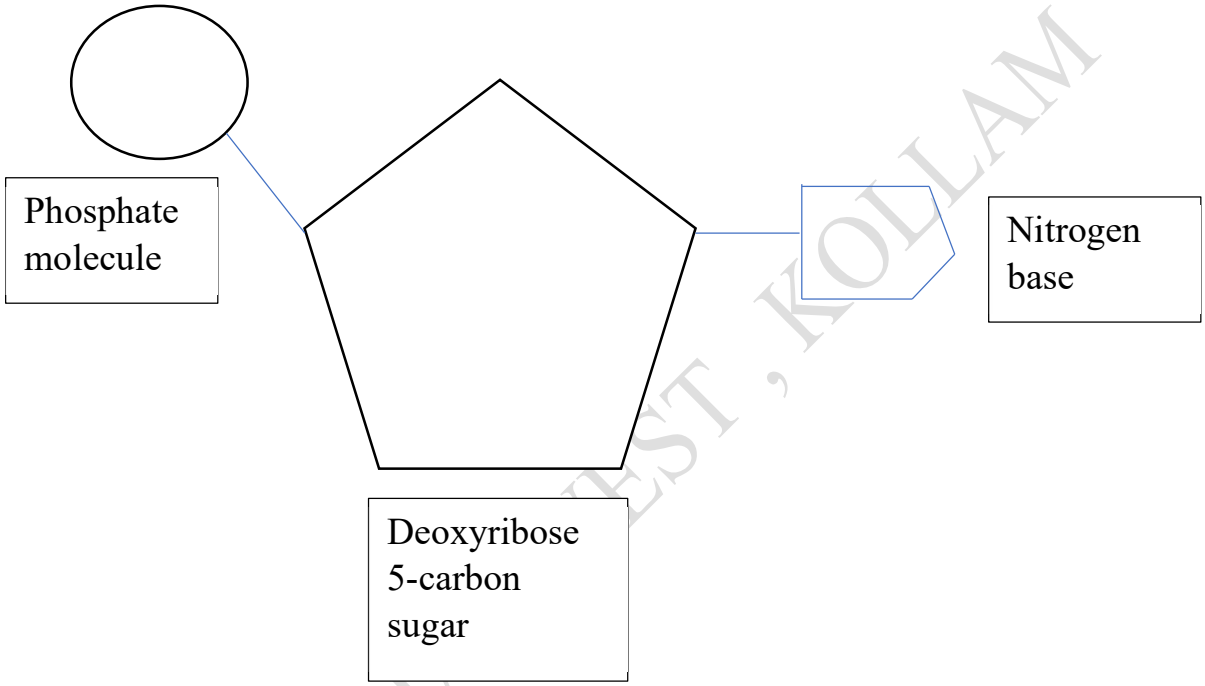
- Adenine always pairs with Thymine
- Guanine pairs with cytosine

WORKSHEET

Number of strands in DNA	
Molecules used to make strands	
Molecules used to make rungs	

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A NUCLEOTIDE – THE BASIC BUILDING UNIT OF DNA



- The DNA molecule is made up of billions of nucleotides linked together.

CHROMOSOMES IN HUMAN BEING

- Somatic chromosomes – 22 pairs (44 no.)

- Sex chromosomes – 1 pair (2 no. X and Y)

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Genetic constitution of male and female

- Male – 44 + XY

-Female – 44 + XX.

RIBONUCLEIC ACID (RNA)

RNA is also made up of nucleotides which are made up of a sugar molecule, a phosphate and nitrogen base.

Nitrogen bases in RNA molecule

- Adenine - Uracil

- Guanine - Cytosine

TYPES OF RNAs

- mRNA - Carries message from DNA
- tRNA - Transfer aminoacid to Ribosome
- rRNA - helps in the formation of bonds between amino acids

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MAIN DIFFERENCES BETWEEN DNA AND RNA

	Number of strands	Type of sugar molecule	Nitrogen bases
DNA	Two	Deoxyribose sugar molecule	• Adenine • Thymine • Guanine • Cytosine
RNA	One	Ribose sugar molecule	• Adenine • Uracil • Guanine • Cytosine

DIFFERENT STAGES OF PROTEIN SYNTHESIS

I. Transcription

- mRNA is formed from DNA where genes are located.

II. Translation

- mRNA reaches ribosome.
- tRNA collect different amino acids and reach ribosome.
- rRNA combine amino acids and make protein.

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Genetics

- Genetics is the branch of science that deals with genes, heredity and variation.
- Gregor Johann Mendel is considered as the father of genetics.

MENDEL'S EXPERIMENTS

MONOHYBRID CROSS

In monohybrid cross only one pair of contrasting traits is considered at a time. Tall and dwarf are a pair of contrasting traits of the character height.

- The different forms of a gene which control a character is called alleles. 'T' and 't' are the alleles of a gene which control the character height

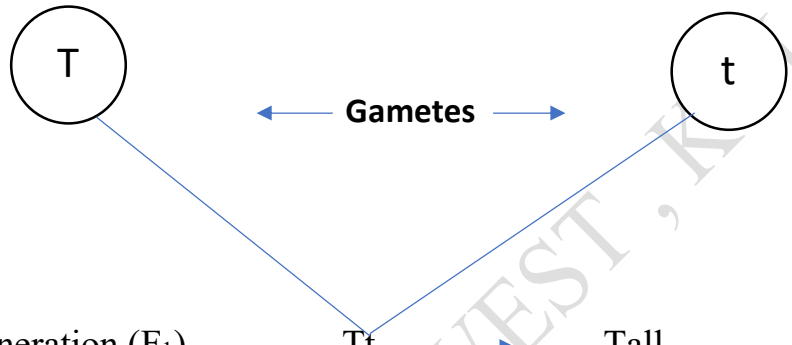
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MONOHYBRID CROSS -ILLUSTRATION

Parental plants

Tall x Dwarf

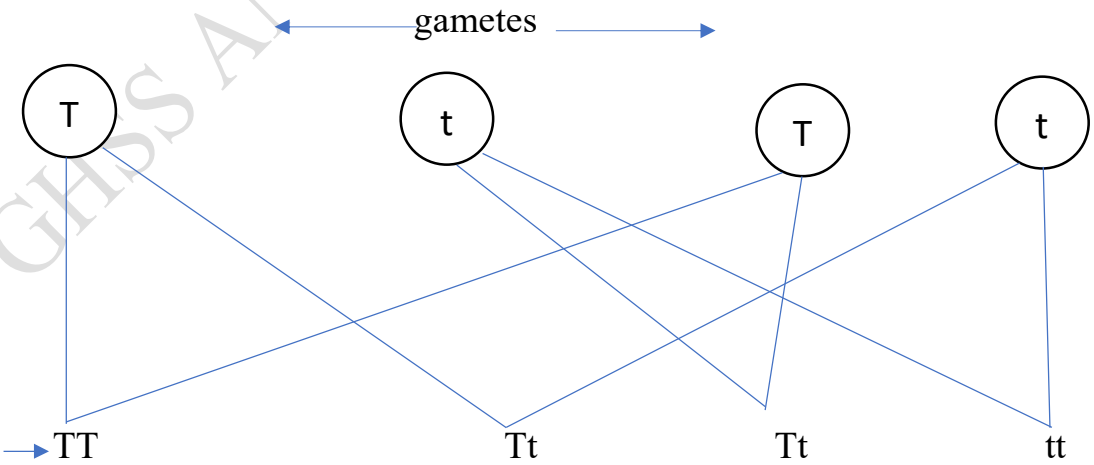
TT tt



First generation (F_1) Tt → Tall

SELF-POLLINATION OF FIRST GENERATION

Tt X Tt



F_2 → TT Tt Tt tt

Tall Tall Tall Dwarf

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MENDEL'S POSTULATES

- A trait is controlled by two factors.
- When a pair of contrasting characters are subjected to cross, only one trait is expressed in the first generation (F_1) and the other trait becomes hidden.
- The trait expressed in F_1 is called dominant trait and the hidden trait is called recessive trait.
- When gametes are formed, the factors that determine a trait separated without mixing.
- In second generation (F_2) the ratio of dominant and recessive trait is 3:1.

DIHYBRID CROSS

Phenotypic ratio - 9:3:3:1

MONOHYBRID CROSS

Phenotypic Ratio - 3:1

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NON MENDELIAN INHERITANCE

Name of Inheritance	Reason	Example
1. Incomplete Dominance	A dominant allele cannot fully hide the allele of the recessive trait.	When a red flowered four o' clock plant is hybridised with a white flowered one, the resulting offspring have pink flowers.
2. Co-dominance	Both alleles exhibit their traits at the same time	Roan coat pattern seen in cattle and horses.
3. Multiple allelism	The condition where more than two alleles found for a gene.	In human, the gene that determine blood group have three alleles. They are I^A,I^B and i.
4. Polygenic Inheritance	A condition in which a trait is controlled by more than one gene.	Skin colour – x in man is controlled by more than one gene.

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GENETIC PROCESSES RESPONSIBLE FOR VARIATIONS

➢ CROSSING OVER

In the first stage of meiosis, homologous chromosomes pair up and exchange genetic material with each other. As a result, recombination of alleles takes place. This results in variation in offspring.

Chiasma:

The point of contact of the paired chromosomes is called chiasma.

➢ MUTATION

The sudden heritable change in the genetic constitution of an organism is called mutation.

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REASONS FOR MUTATION

- Errors during DNA replication
- Due to certain chemicals
- Due to radiations, etc.

changes in allele combinations during fertilization, crossing over, mutation etc, cause variations in organisms.

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CHAPTER 2

PATHS OF EVOLUTION

1. THEORIES OF EVOLUTION

a. Lamarckism (Theory of inheritance of Acquired characters)

b. Darwinism

c. Neo Darwinism

a) LAMARCKISM – JEAN BAPTISTE LAMARCK

- Change in environment
- Use and disuse of organs
- Acquired characters
- Inheritance of acquired characters
- Evolution

LIMITATION OF LAMARCKISM

- Acquired characters do not bring change in the genetic structure of an organism and therefore, are not inherited

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b) DARWINISM – CHARLES DARWIN

- **Over production :**

  Every species produces more number of offsprings than can survive on Earth

- **Variation:**

  In organisms favourable or unfavourable variations are formed

- **Struggle For Existence**

  Organisms struggle or compete with each other for food and shelter

- **Survival of the fittest**

  Organisms with favourable variations survive in the struggle for existence

- **Natural selection**

  Favourable variations are inherited and accumulated over generation and new species will originate

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LIMITATION OF DARWINISM

Darwin had no idea about genetic basis of variation and inheritance

c) NEO DARWINISM

- The causes of variations that lead to evolution were genetic changes, genetic recombination and gene flow
- The evidences and further studies from the fields of population genetics, palaeontology, environmental science etc. were added to Darwinism

2. MODERN RESEARCHES THAT TAKE PLACE IN THE FIELD OF EVOLUTION

- Evolutionary clinical medicine
- Personalized medicine
- DNA studies &AI

EVOLUTIONARY CLINICAL MEDICINE

- It studies how bacteria or viruses evolve overtime to become resistance to drugs
- This lead to the creation of new methods for treatment

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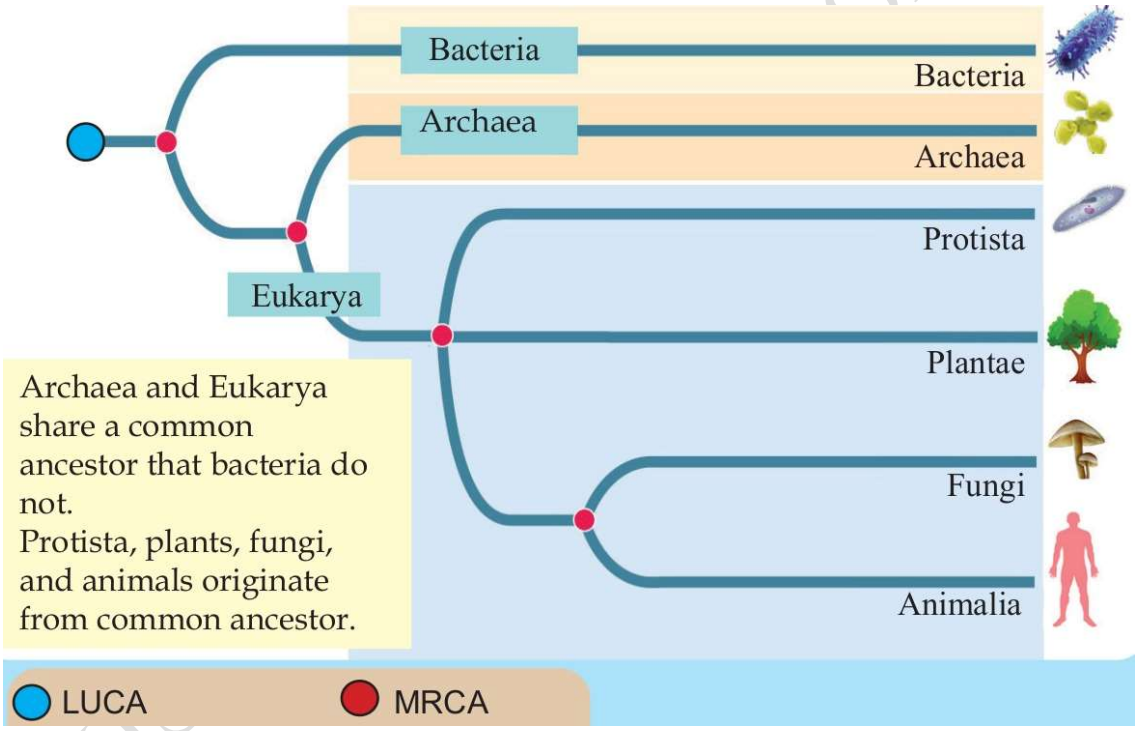
PERSONALIZED MEDICINE

It is designed by looking at an individual's gene and family genetic history

3. EVOLUTION OF DIFFERENT CATEGORIES OF ORGANISMS

LUCA - Last Universal Common Ancestor

MRCA - Most Recent Common Ancestor



4. SPECIATION

The process in which new species arise from a common ancestor is called speciation

5. EVIDENCES OF EVOLUTION

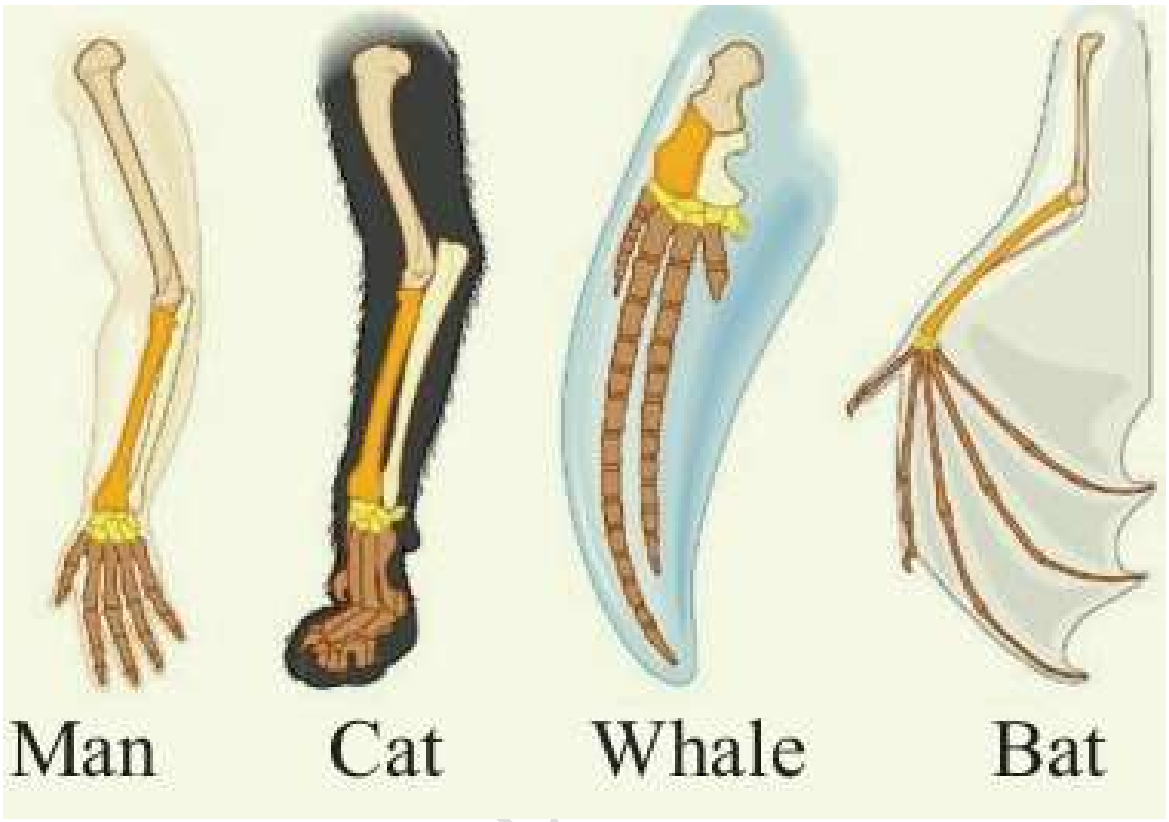
a) MOLECULAR BIOLOGY

- The evolutionary relationship of organisms can be found out by comparing the sequence of nucleotides in the DNA and the sequence of amino acids in proteins in any organism with those of other organisms.

Organism	Difference in the amino acids of the beta chain as compared to humans
Chimpanzee	0
Gorilla	1
Rat	31

- Chimpanzee has the closest evolutionary relationship with humans because there is no difference in the amino acids of the beta chain as compared to humans.

b) COMPARATIVE ANATOMY



- The bones in the forelimb of humans and cats, the flippers of whales and the wings of bats are similar.
- These organs differ in their external structure and functions.

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c) FOSSIL EVIDENCES

- Organic evolution is a gradual process.

  Eg: The ancestors of horses had shorter legs than the ones in our times.

- Fossils which are connecting links reveal the evolutionary relationship between organisms.

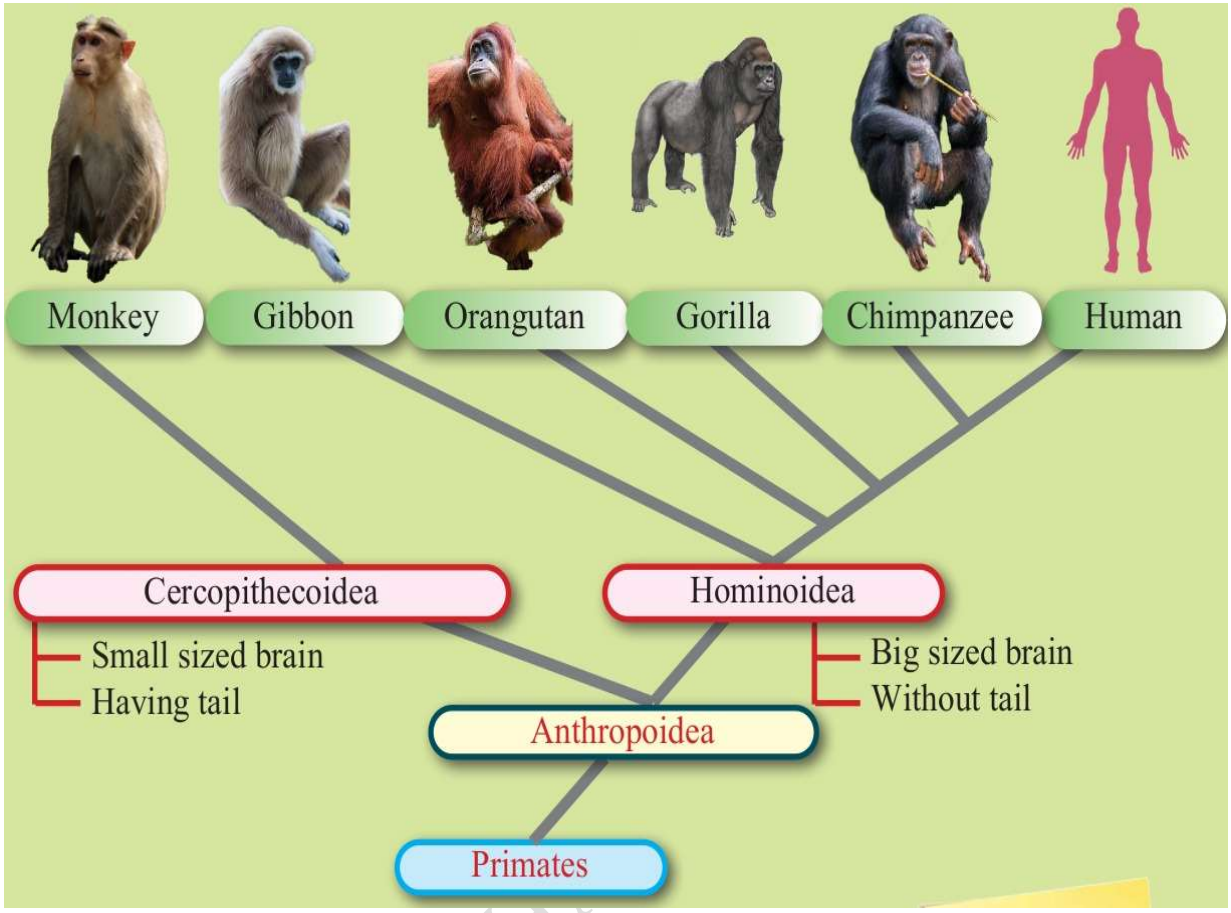
  Eg: Archaeopteryx.

- It is proven that many species have become extinct that were living on Earth in the past.

  Eg: Dinosaurs, Mammoths.

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6. HUMAN EVOLUTIONARY TREE



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The common characteristics of Primates:

- An opposable thumb
- Binocular vision
- Well-developed brain
- Social behaviour and communication skills

7. MEMBERS IN THE HUMAN EVOLUTIONARY PATH

Humans	Cranial capacity	Characteristic
Sahelanthropus tchadensis	350 cm^3	First link in the human evolutionary series
Australopithecus	450cm^3	Bipedalism
Homo habilis	600cm^3	• Large skull • Made tools with stones • Began hunting
Homo erectus	900cm^3	• Walk upright on two legs • Large forehead • Omnivores • Used stone weapons for hunting
Homo neanderthalensis	1450cm3	• Contemporaries of modern man • Small, sloping forehead

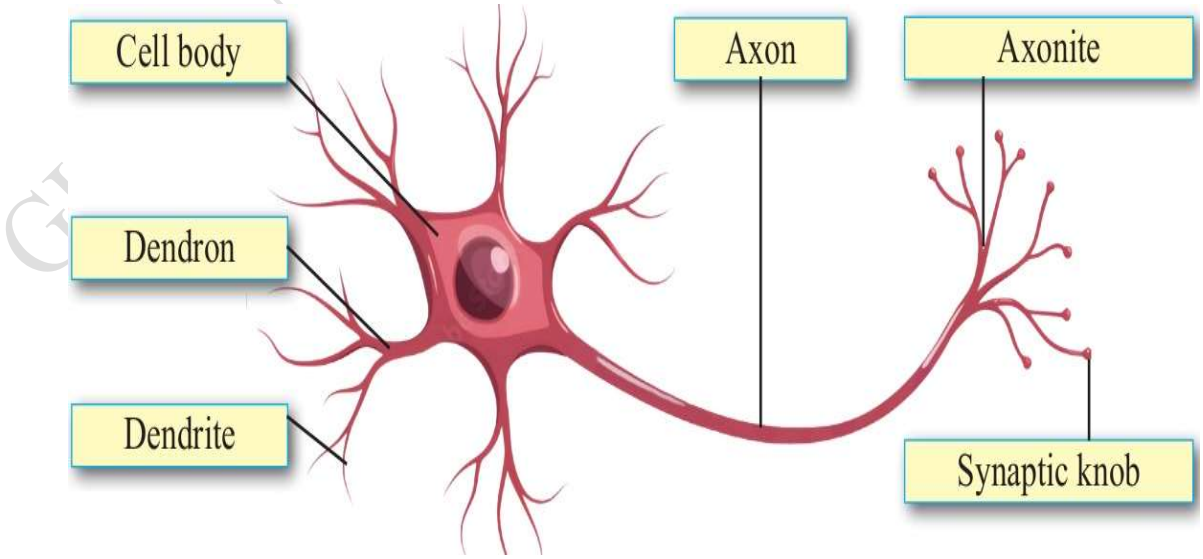
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		• Thick eye brows • Buried dead bodies
Homo sapiens	1350cm3	• Modern man • Acquired technology and agricultural methods • Built cities • Domesticated animals

8. HUMAN NERVOUS SYSTEM

NEURONS

Neurons are the basic building blocks of nervous system



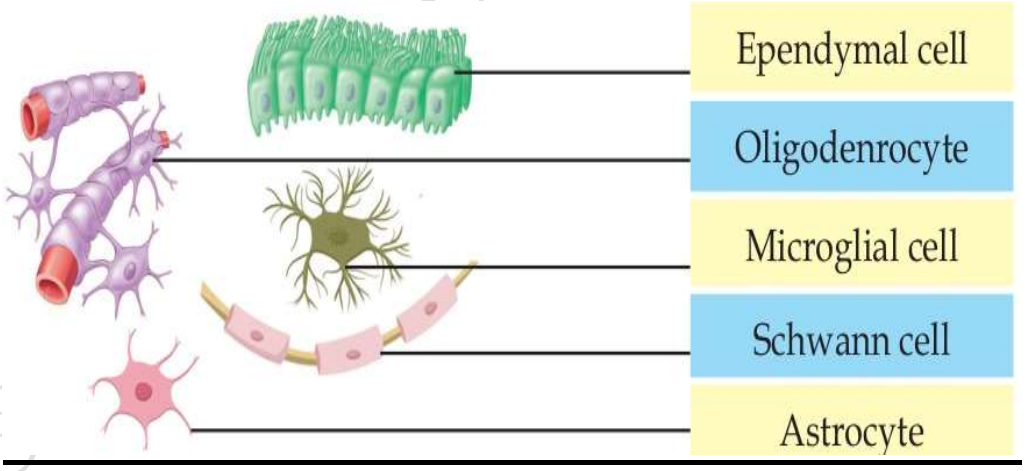
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Parts of neuron	Peculiarities	Functions
Cell body	Centre of the neuron	
Dendron	Short filament from the cell body	Carries impulses from dendrites to the cell body
Dendrite	Branches of dendron	Part that receives impulses from adjacent neurons
Axon	Longest filament from the cell body	Carries impulses from cell body to outside
Axonite	Branches of axon	Carries impulses to synaptic knob
Synaptic knob	Tip of axonite	Secretes neurotransmitter

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9. NEUROGLIAL CELLS

Peculiarities	Function	Examples
• Seen in brain and spinalcord • It has the ability to divide • They cannot receive stimuli or transmit messages	• Bring nutrition necessary for neurons • Eliminate waste • Act as defense cells	• Ependymal cell • Oligodendrocyte • Microglial cell • Schwann cell



10. MYELIN SHEATH

- The axon is covered by a layer called Myelin sheath
- It is made up of shiny white fat called myelin.

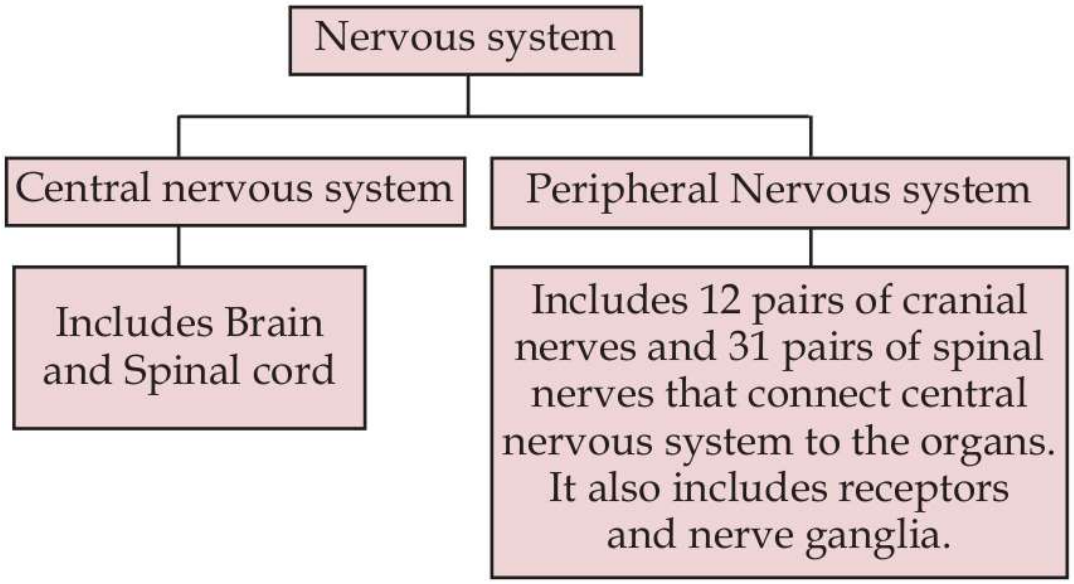
FUNCTIONS

- Act as an insulator
- To provide nourishment to the neuron
- To protect axon from external injuries

WHITE MATTER AND GREY MATTER

- The part where the abundant myelinated neurons in the brain and the spinal cord is called white matter
- The part where the cell bodies and parts of the neurons without a myelin sheath are seen is called grey matter.

11. PARTS OF NERVOUS SYSTEM



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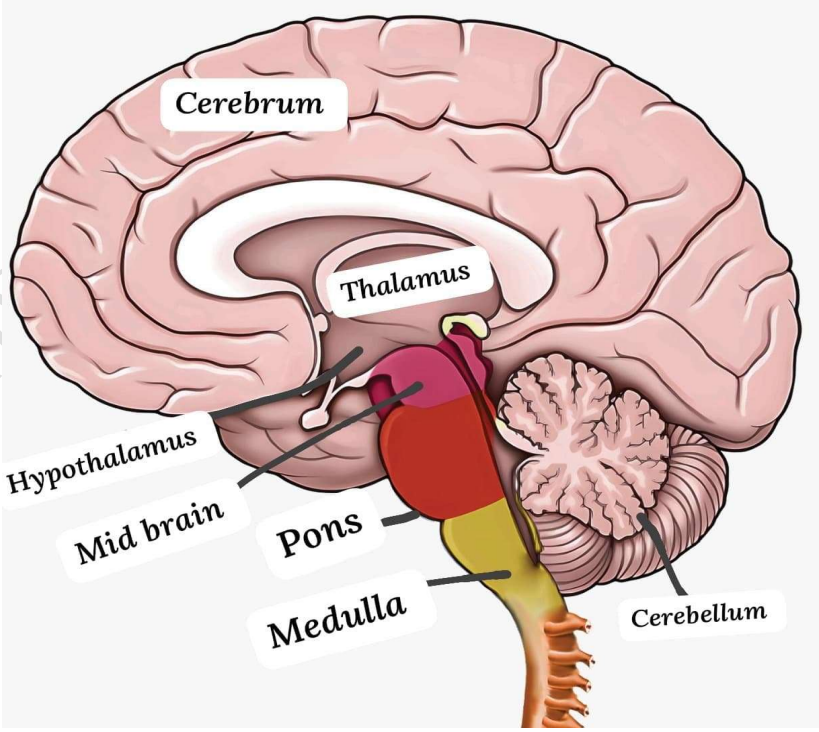
PROTECTION OF NERVOUS SYSTEM

- Three-layered meninges
- Cerebrospinal fluid

FUNCTIONS OF CEREBROSPINAL FLUID

- Provides oxygen and nutrients to the tissues
- Eliminates waste
- Regulates the pressure
- Protects from external injuries

PARTS OF BRAIN AND THEIR FUNCTIONS



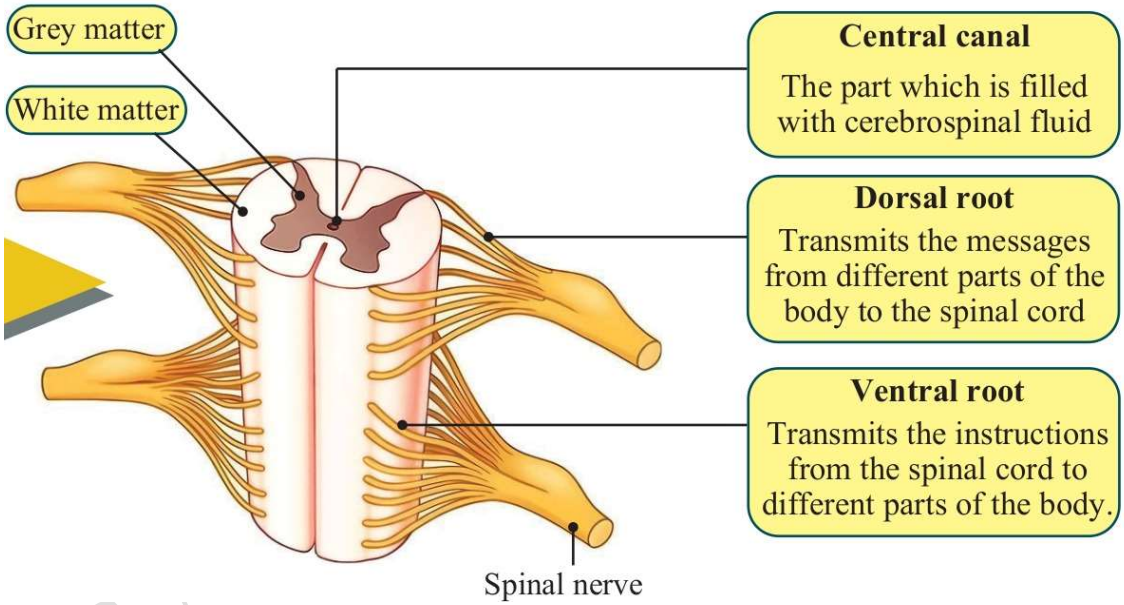
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Parts	Functions
Cerebrum	• Centre of memory, intelligence, thinking and imagination. • It has the role in problem solving, planning and voluntary movements.
Cerebellum	• Maintains equilibrium of the body by coordinating muscular activities.
Thalamus	• Acts as a relay station of messages to and from the cerebrum.
Hypothalamus	• Regulates body temperature, hunger, thirst and emotions. • Maintains homeostasis.
Midbrain	• Assessment of messages regarding vision and hearing. • Plays a role in the movement of eyes and eyebrows.
Pons	• Coordinates the muscular activities of the eye and face.

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	• Regulates the rate of ventilation.
Medulla oblongata	Controls involuntary activities like heartbeat, ventilation, vomiting, cough, sneezing, etc.

12. STRUCTURE OF SPINAL CORD



FUNCTIONS OF SPINAL CORD

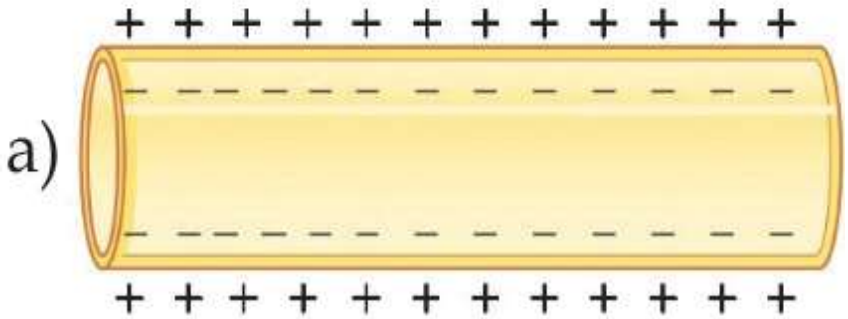
- Transmits messages from different parts of the body to the brain.

- Transmits instructions from the brain to different parts of the body.

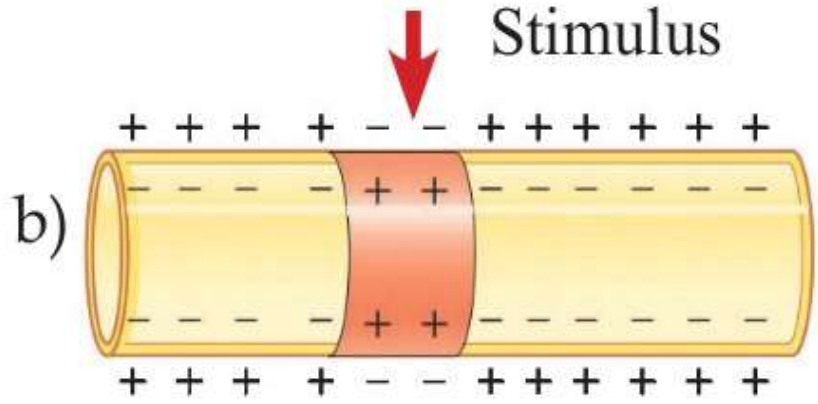
13. TRANSMISSION OF IMPULSES

- The messages transmitted through the neuron are called nerve impulses.

a) The outer surface of the cell membrane is positively charged and charged when not stimulated.

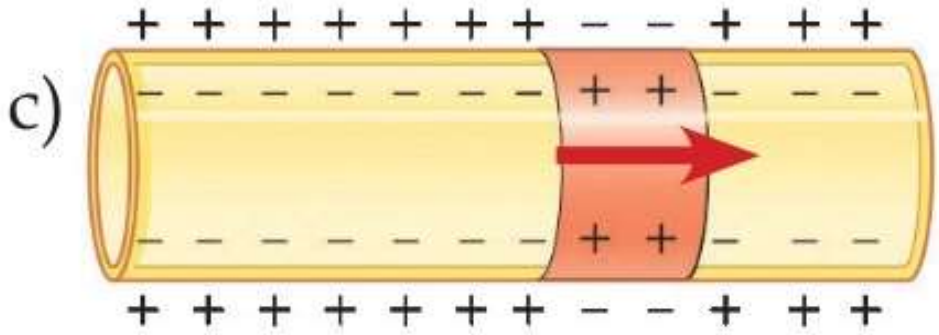


b) The charge distribution reverses at the point of stimulus. The outer surface becomes negative and the inner surface becomes positive at that spot.



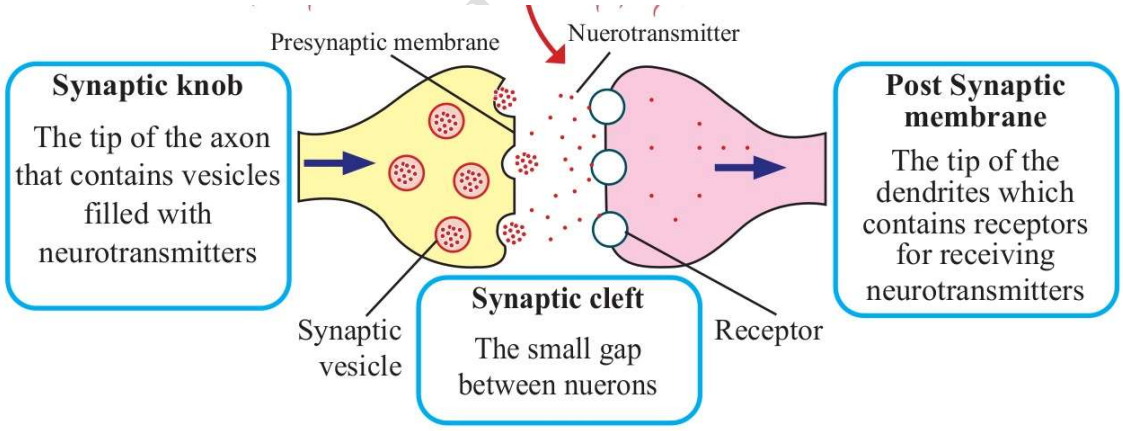
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c) The reversed charges move forward, causing the impulse to propagate along the axon.



14. SYNAPSE

- The part where an impulse is transferred from one neuron to another is called synapse.



FUNCTIONS OF SYNAPSE

- Increases the speed of impulses
- Transmits the impulses in only one direction

15. DIFFERENT TYPES OF NEURONS

Neurons	Functions
Sensory neuron	Carries information from sensory receptors to the central nervous system
Motor neuron	Transmits impulses from the central nervous system to various parts of the body
Interneuron	Relays and processes signals between sensory and motor neurons

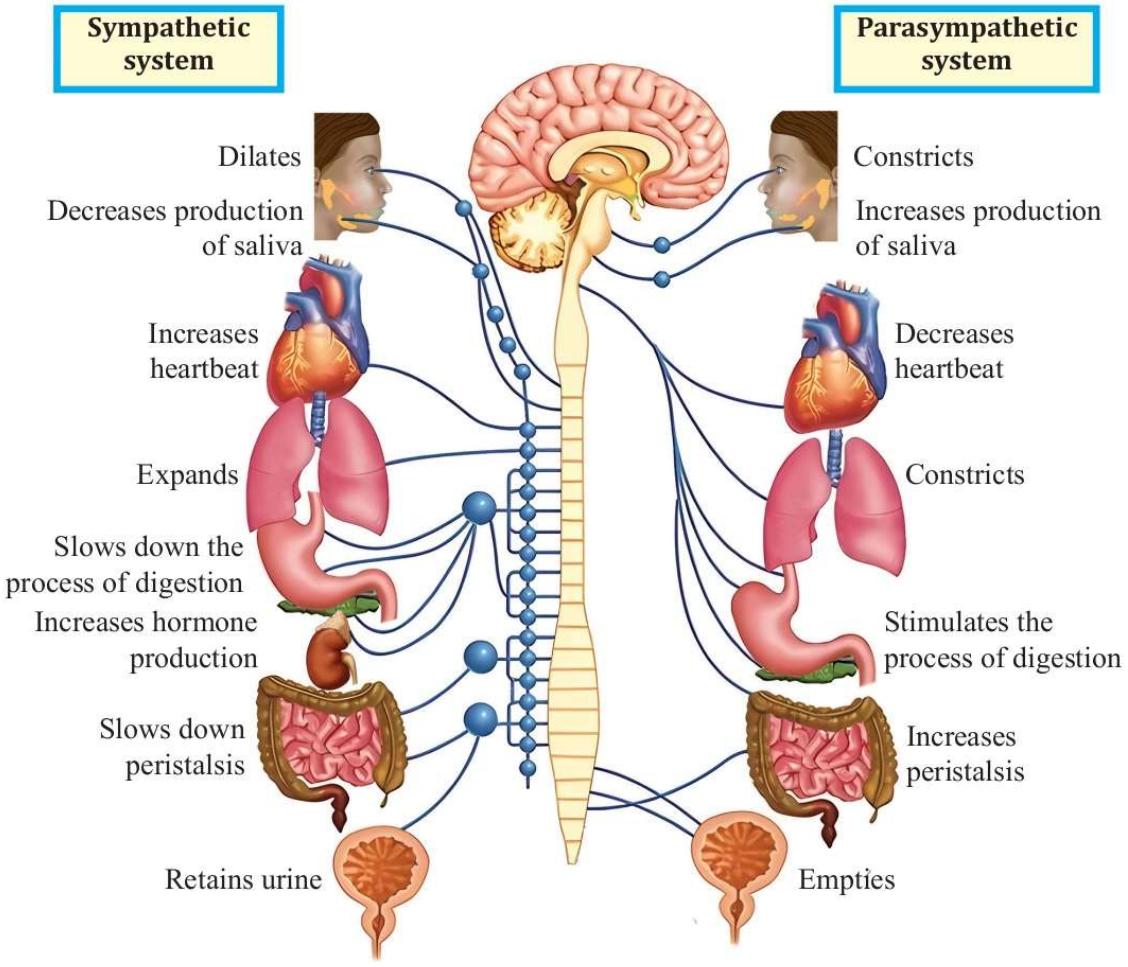
16. DIFFERENT TYPES OF NERVES AND THEIR FUNCTIONS

Nerve	Building Block	Function
Sensory nerve	Sensory neuron	Transmits instructions from the organs to the central nervous system
Motor nerve	Motor neuron	Transmits instructions from the central nervous system to the organs

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Mixed nerve	Both sensory neuron and motor neuron	Carries both sensory impulses to the central nervous system and motor impulses from the central nervous system to muscles or glands

17. AUTONOMOUS NERVOUS SYSTEM

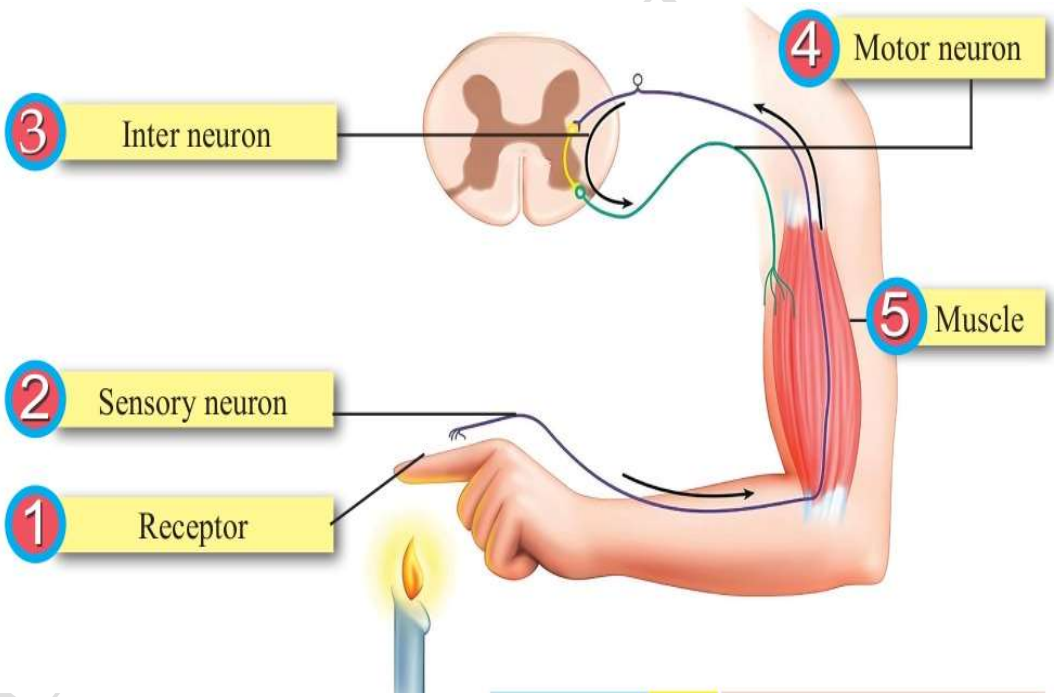


18. REFLEX ACTION

Reflex actions are reactions that occur spontaneously and involuntarily in response to stimuli.

REFLEX ARC

- The pathway through which impulses are transmitted in a reflex action is called reflex arc



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THE NERVOUS SYSTEM OF DIFFERENT ORGANISMS

Hydra	A neural network with no control center is seen.
Planaria	A pair of nerve ganglia in the head region.
Insects	The neurons in the head region unite to develop a brain.

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CHAPTER III

BEHIND SENSATIONS

STIMULI:

- Stimuli are the circumstances that lead to responses in living beings

RECEPTORS:

- Stimuli are recognised by the body through Specialized cells or nerve endings. They are known as Receptors.

BLIND SPOT:

- There are no photoreceptor cells in retina where the optic nerve originates. This part having no vision is known as the blind spot.

YELLOW SPOT (MACULA)

- The part of the retina where cone cells are abundant

MELANIN.

- The pigment present in Iris is called melanin.

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- Melanin gives the Iris its characteristic Colour as well as absorbs ultraviolet rays

CRYSTALLIN

The main structural Component of the lens is a protein Called Crystallin

LIGAMENTS

The part which fixes lens to ciliary muscles

POWER OF ACCOMMODATION

The ability of the eye to focus images of both near and distant objects accurately on the retina is called power of accommodation

RHODOPSIN

The pigment present in Rod cells

PHOTOPSIN

The pigment present in cone cells..

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BINOCULAR FUSION

The images formed in the retina of two eyes are sent to the visual cortex in the brain. The process of comparing the images and fusing the images that reaches in the brain is called Binocular fusion.

EUSTACHIAN CANAL

- The eustachian canal is a long tube that is 4 cm long that connects middle ear to pharynx.
- It helps to balance the pressure on both sides of the tympanum

ORGAN OF CORTI

- The part of ear where auditory receptors are present is called organ of corti
- It present in the basilar membrane between middle and lower chambers of cochlea.

VESTIBULAR SYSTEM

The part of internal ear related to body balancing

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EYESPOT (STIGMA)

The photoreceptors present in euglena

OMMATIDIA

The compound eye present In Insects

ANTENA

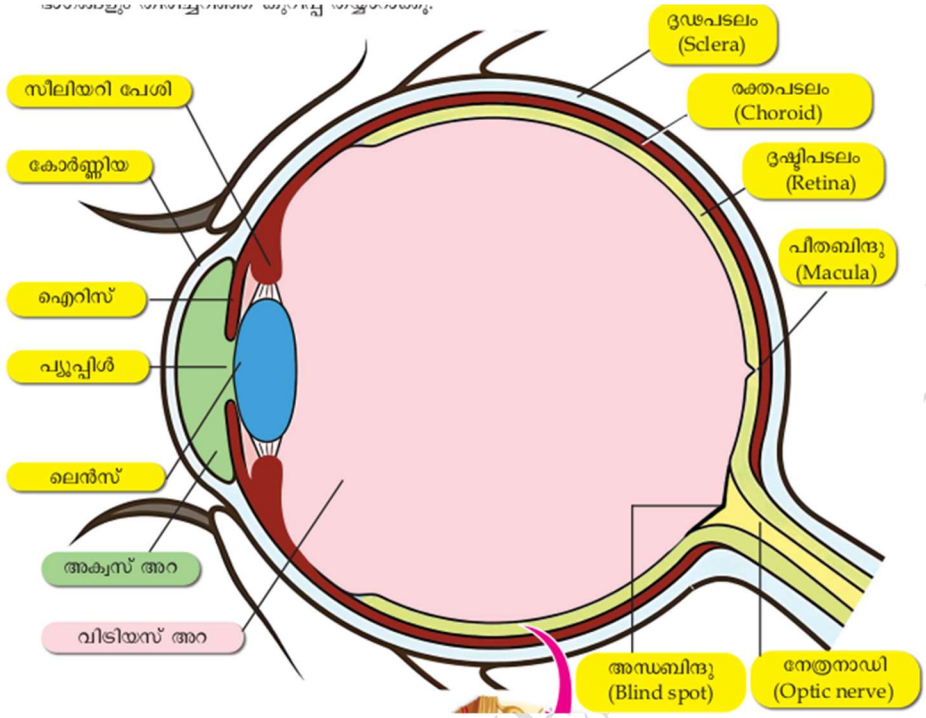
The part which helps to detect Smell and touch.

SENSE ORGANS, RECEPTORS AND STIMULI

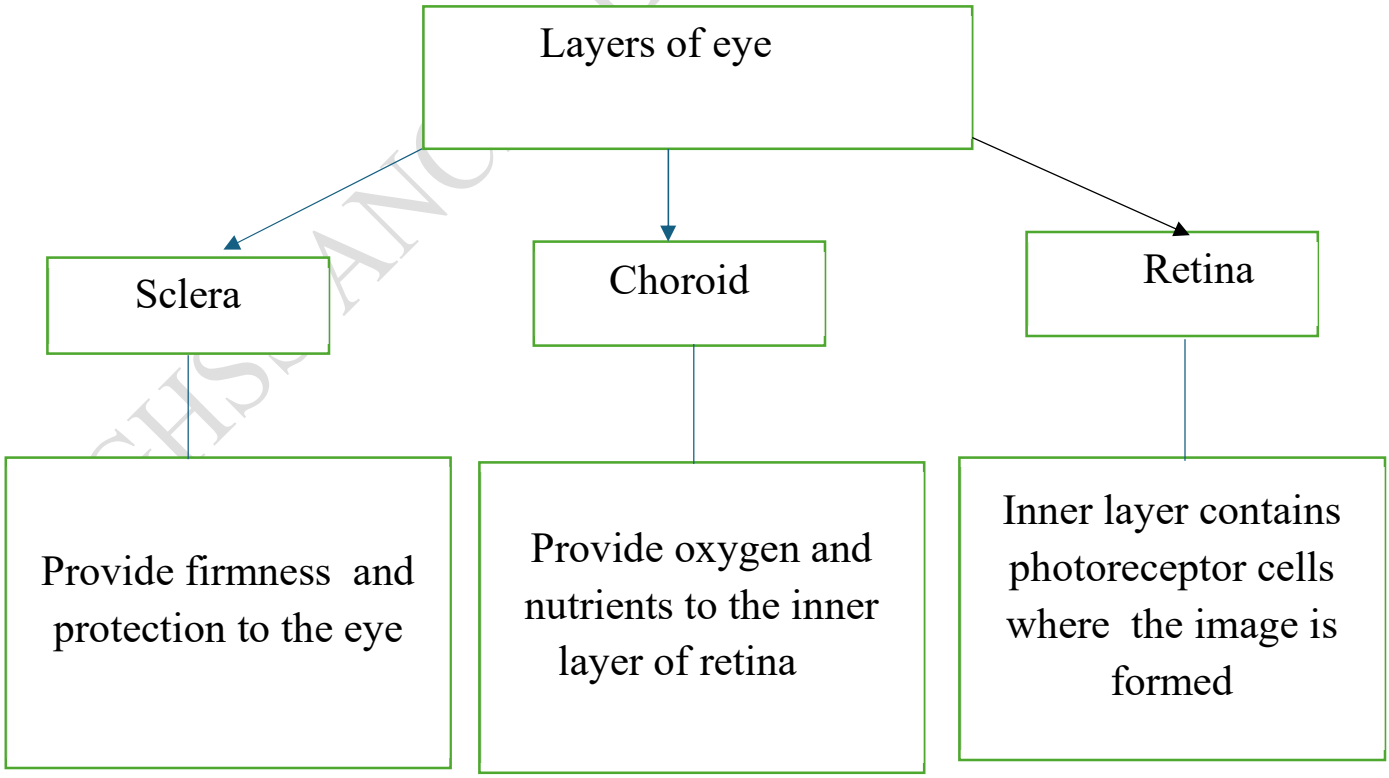
Sense organs	Receptors	Stimulus
Eye	Photoreceptors	Light
Ear	Audio receptors	Sound
Tongue	Chemo receptors	Taste
Skin	Touch Receptors	Touch
Nose	Olfactory receptors	Smell

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STRUCTURE OF THE EYE



LAYERS OF EYE



PART OF SCLERA

Cornea - The transparent anterior part of the eye

PART OF CHOROID

1. **Ciliary muscles** - Adjusts the curvature of lens.
2. **Iris** - 2 types of muscles in the iris regulate the Size of pupil
   - Regulate the amount of light.
   - Melanin a pigment present in Iris
3. **Convex lens**

   Forms a Small, real and inverted image of the object on the retina

PART OF RETINA

1. **Layer of photoreceptors**

   Rod cells and cone cells are present

2. **Bipolar cell layer**

   Transmits impulses from photoreceptors to ganglion cells.

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3. Ganglion cell layer

Transmits impulses from bipolar cells to the optic nerve.

EYE CHAMBERS AND HUMORS

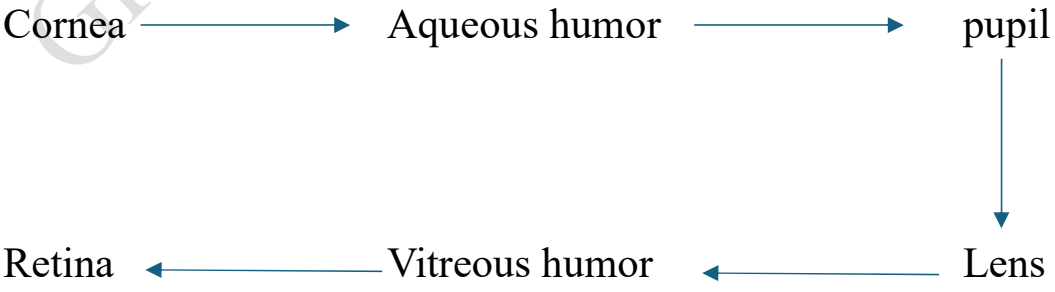
Aqueous chamber.

- The chamber is seen between the cornea and lens.
- The watery aqueous humor is present here

Vitreous Chamber

- The chamber lies between the lens and retina
- The transparent jelly like Vitreous humor present there maintains the shape of eyeball

The path of light rays through eye



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THREE TYPES OF CONE CELLS

S-Cones - Having short wavelengths Blue light

M - Cones - Medium Wavelength (Green light)

L -Cones - Longer Wavelength (Red light)

- The gene Responsible for the production of Pigments in Cone cells which are sensitive to green and red are found in x chromosome

The gene which is responsible for the production of blue cone pigment IS found in Chromosome number 7

MUSCULAR ACTIVITY IN PUPPL

Intense light	Circular, muscle	Contracts
Dim light	Radial muscle	Contracts

THE MUSCLES RELATED TO PUPIL

- Circular Muscle
- Radial Muscle

THREE PARTS OF LENS

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- Capsule
- Lense. fibres
- Epithelium

MUSCLE RELATED TO LENS

- Ciliary Muscle

EYE DISEASE / DISORDERS

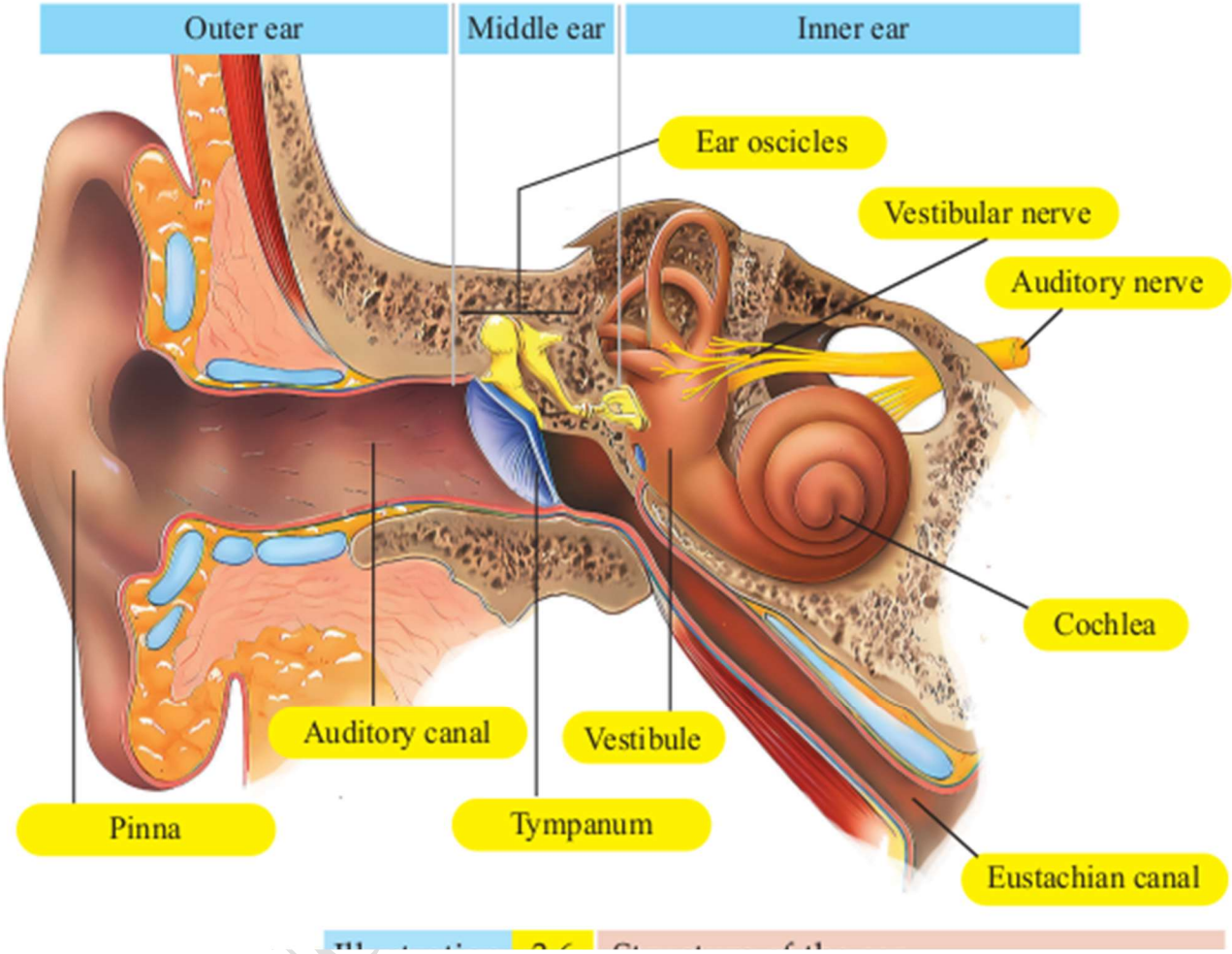
Eye disease /disorders	Reason	Treatment
Short sightedness	Enlarged Eyeball	Spectacles with Concave lens or contact lens surgery
Long sightedness	Shortened Eyeball	Spectacles with Convex Lens, contact lens, surgery
Astigmatism	Defect in curvature of lens or cornea	Cylindrical lens

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Cataraet	Eye lens becomes opaque	Surgery
Glaucoma	• Failure in the reabsorption of aqueous humor. • Increased pressure in the eye	Early treatment
Conjunctivitis	Infection in the conjunctiva	Antibiotic treatment, eye hygiene
Diabetic retinopathy	Uncontrolled diabetes	Diabetes control
Night blindness	Deficiency of Vitamin A	Consume Vitamin A rich food
Xerophthalmia	Due to prolonged deficiency of Vitamin A, Cornea becomes opaque	Consume Vitamin A rich food

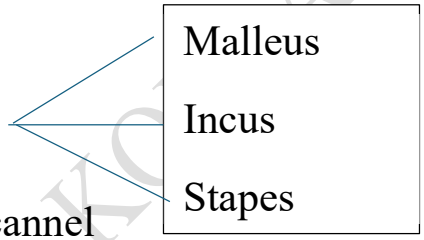
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STRUCTURE OF EAR



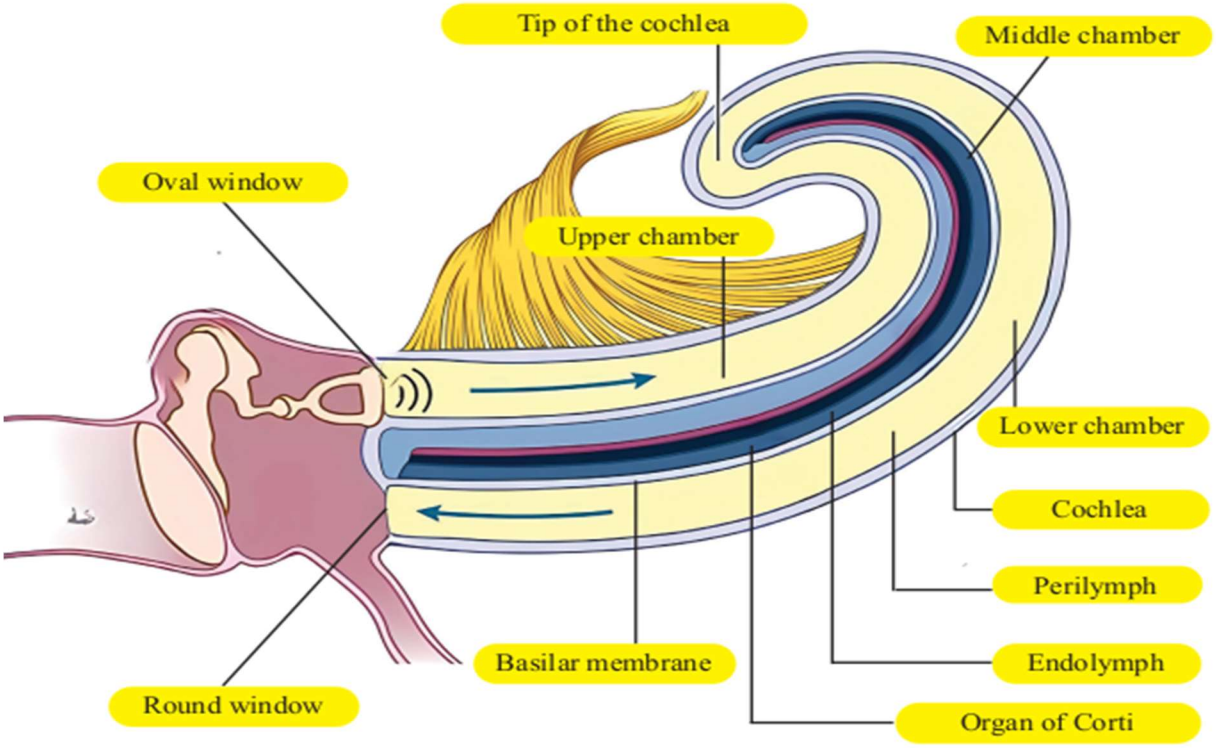
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Main parts of the ear	Associated parts
Outer Ear	• Pinna • Auditory canal
Middle Ear	• Tympanum • Ear Ossicles • Eustachian cannel
Internal Ear	• Vestibule • Cochlea • Semicircular canals • Auditory nerve • Vestibular nerve



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INNER EAR COCHLEA



PARTS OF EAR AND THEIR FUNCTIONS

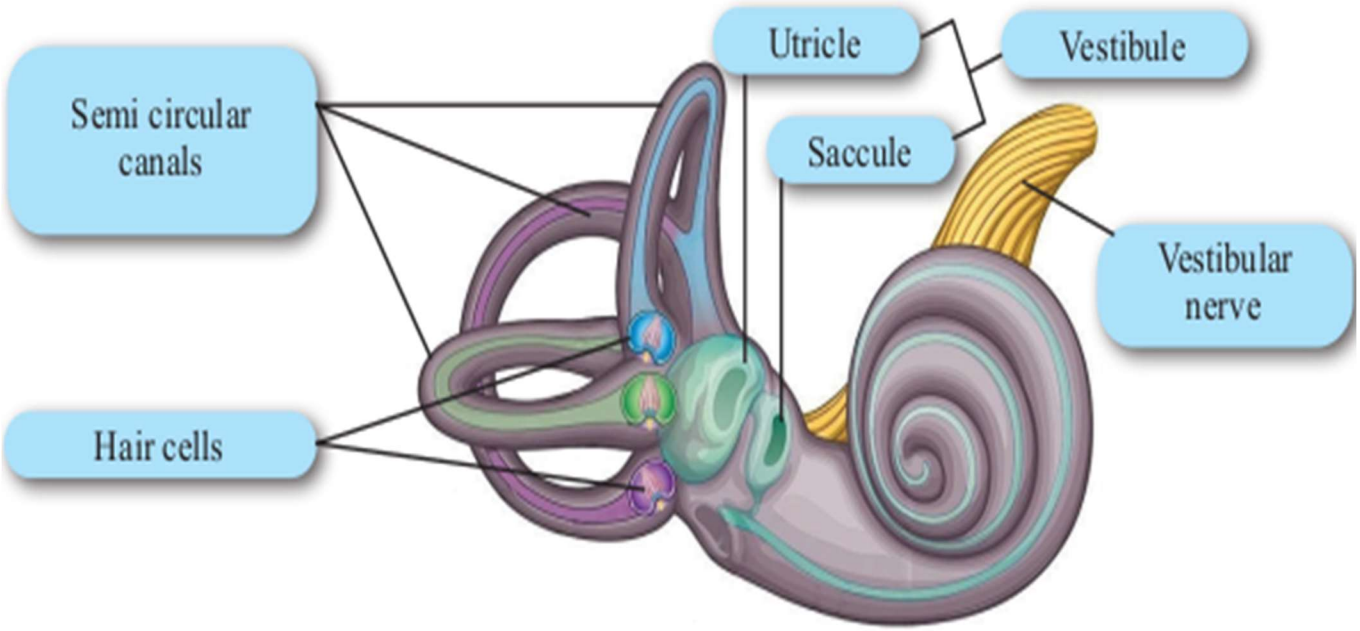
Pinna	• Direct sound waves into auditory canal • Identify the direction of sound wave • Protect auditory canal
Auditory canal	• Directs sound waves to the tympanum • Protects tympanum from foreign particles

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	• Ear wax and sebum present in it
Ear ossicles	Amplify vibrations and passes it to the ossicles
Eustachian canal	• Connect middle ear to pharynx • Balance the pressure on both sides of tympanum • Facilitate flow of mucus and fluid from middle ear to the pharynx
Oval window	• Vibrates the fluid, perilymph and there by vibrations in endolymph
Cochlea	• Contain auditory receptor in organ of corti in the basilar membrane, in the cochlea • Contain perilymph and endolymph
Vestibular system	• Vestibule • 3semicircular canals
Auditory nerve	Send impulses from • Cochlea to cerebrum and • Vestibular system to cerebellum

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Inner ear – body balancing



SENSE OF HEARING

Sound → Pinna → Auditory Canal →

Tympanum → Malleus, Incus, Stapes → Oval Window

→ Cochlea → Basilar Membrane →

Organ of Corti → Impulse → Auditorynerve →

Cerebrum → hearing

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BODY BALANCING FLOW CHART

Movement in Head ⟶ Hair in Vestibular system ⟶ Impulses

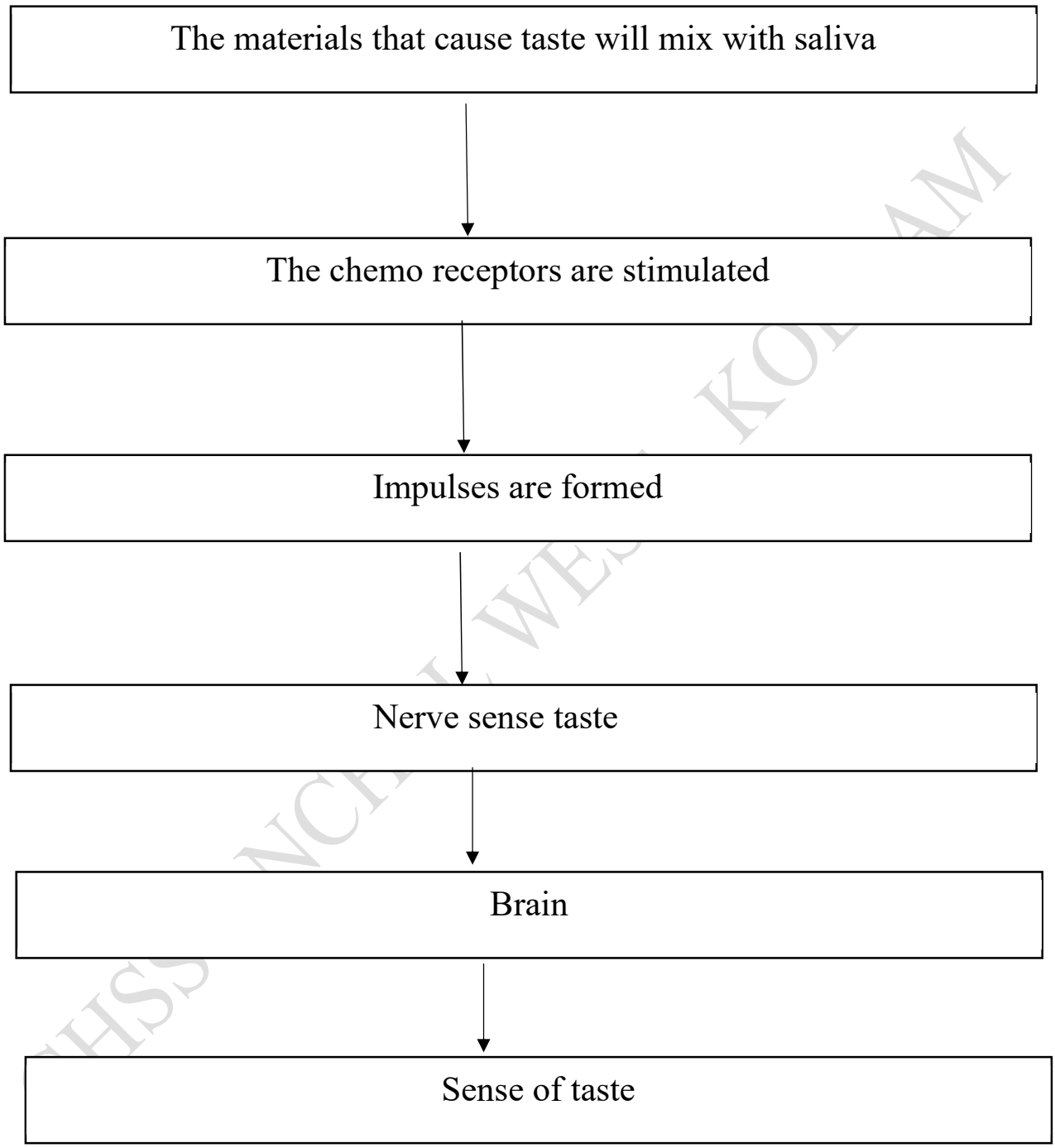
Vestibular Nerve ⟶ Cerebellum ⟶ Body Balancing

SENSORY DIVERSITY IN ORGANISMS

Amoeba	Movement according to the pressure of chemicals surrounding it
Bat	Echolocation organ
Hawk	Eye with high vision system detecting UV rays
Dog	Highly sensitive olfactory receptors
Insects	Ommatidia
Snake	Jacobson's organ to detect smell
Insects	Ommatidia
Euglena	Eye spot, a photo receptor is seen

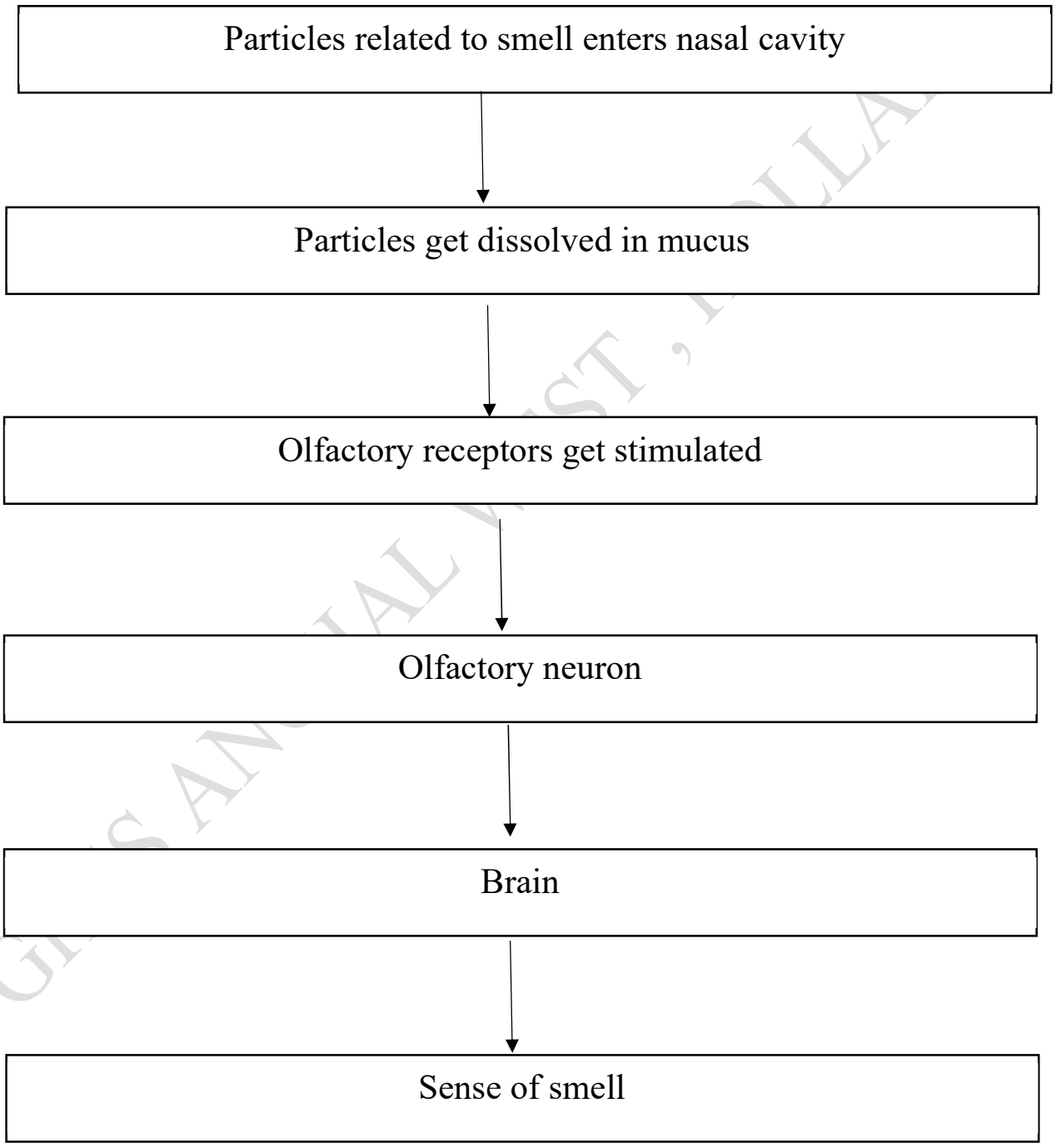
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FLOW CHART RELATED TO TASTE



HSS NCERT WEB KOLLAM

FLOW CHART RELATED TO SMELL



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RECEPTORS AND EXPERIENCE

Process	Smell	Taste
The fluid in which substances dissolve	Mucus	Saliva
Stimulated receptor	Olfactory Receptor	Chemo Receptor
The nerve that carries Impulse to Brain	Olfactory nerve	Nerve that sense taste

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CHAPTER IV

CHEMORECEPTION IN ORGANISMS

F. W WENT: A botanist from Holland Cut the coleoptile of the oats plant while Studying the growth of plants, and kept it in a jelly like substance called agar

TARGET CELL

- A cell in which a hormone acts is called target cell
- Receptors are seen in the cell membrane of target cell
- Hormone combine with their receptors to form a hormone receptor complex

PRODUCTION OF HORMONES

In Animals - Hormones are produced by endocrine glands.

In Plants - Hormones are produced at the tip of shoot or root, in seeds or in sprouting leaves.

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PLANT HORMONES AND THEIR FUNCTIONS

PLANT HORMONES	FUNCTION
Auxins	• Increase the length of the stem • Regulate tropic movements • Stimulate the formation of flowers and fruits
Gibberellins	• Break seed dormancy • Influence the growth of fruits and seeds • Stimulate cell division on both stems and roots
Cytokinins	• Stimulate cell division and cell differentiation in germinating seeds • Stimulate the growth of lateral buds • Delay senescence and retains the green color of leaves
Abscise acid	• Maintains the dormancy of seed • It promotes abscission of mature leaves and fruits.
Ethylene	• Degradation of chlorophyll, proteins and nucleic acid leading to the ageing

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	of leaves and flowers and ripening of fruits

SYNTHETIC PLAT HORMONES AND THEIR USES

SYNTHETIC PLANT HORMONES	USES
Synthetic auxins	• Used to induce root formation in stem cuttings • Used as weedicide (2,4-D)
Synthetic Gibber lines	• To increase the size of fruits in grapevines • Promote in stem elongation in sugarcane
Synthetic Ethylene	• Used to make fruits ripen uniformly for harvest at the same time • Ethylene is used to ripen bananas and tomatoes simultaneously

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PHOTOPERIODISM

Flowering is closely related to the length of day and night. This response of plants to photoperiod is known as photoperiodism.

Phytochrome - A Light sensitive pigment plays a major role in photoperiodism

THYROID GLAND

Position - Below the Larynx

Cells - two types of cells are present

1) Follicular cells
2) Parafollicular cells

Hormones - Two Hormones

1) Thyroxine (Produced by Follicular cells)
2) Calcitonin (Produced by Parafollicular cells)

FUNCTION

- Metabolism
- Body Growth development

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THYROID HORMONES AND THEIR FUNCTIONS.

Hormones	Functions
Thyroxine	• Control metabolism and body temperature • Development of brain in newborn babies and children • Helps to maintain the health of heart, skin reproductive system
Calcitonin	Reduce the level of calcium in blood

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DEFECTS RELATED TO THYROID GLAND

Condition	Causes	Symptoms
Hyperthyroidism	Excessive secretion of thyroxine	• Increase the rate of metabolism • Weight loss • Heart beat increases • Excessive sweating anxiety

Hypothyroidism	Reduced secretion of thyroxine	• Metabolism decreases • Body weight increases • Heart rate slows down • Fatigue • Intolerance to cold
Goiter	Various disorders of thyroid gland, deficiency of iodine	• Swelling in neck region • Difficulty in swallowing

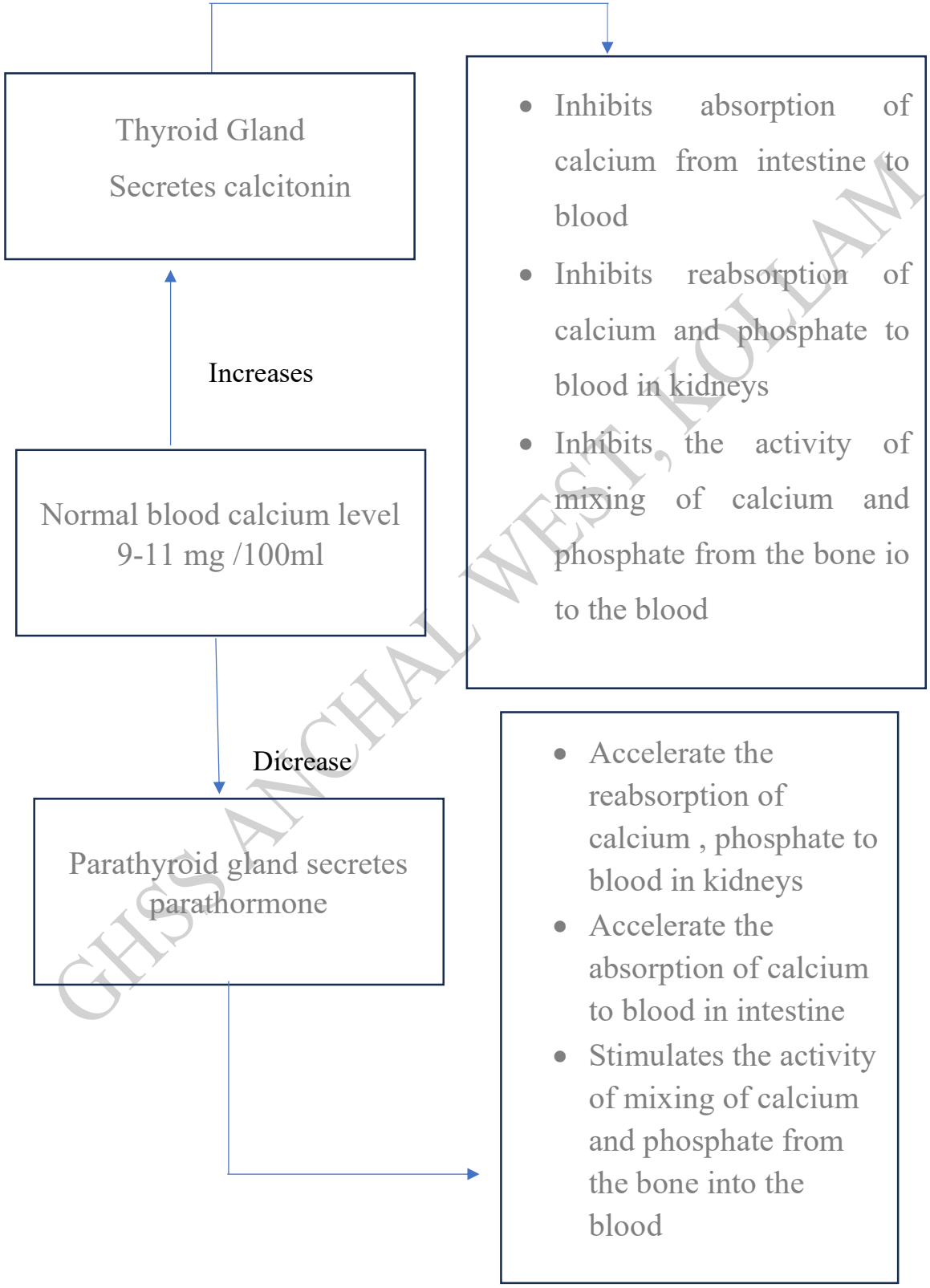
PARATHYROID GLAND

Position : Parathyroid gland located behind thyroid gland

Hormone : Parathormone

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REGULATION OF CALCIUM IN BLOOD



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PANCREAS

Position : Seen near the initial region of intestine

Cells : A group of cells in pancreas is called Islets of Langerhans, functions as endocrine gland

Two type of cell in Islets of Linertrains

- Alpha cells
- Beta cells

Hormones

- Insulin (secreted by Beta cells)
- Glucagon (secreted by Alpha cells)

DIABETES MELLITUS

The level of glucose in blood rises above normal (FBS above 126 mg/dl)

SYMPTOMS

- Excessive thirst
- Hunger
- Frequent urination

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- Weight loss
- Fatigue

REGULATION OF GLUCOSE IN BLOOD

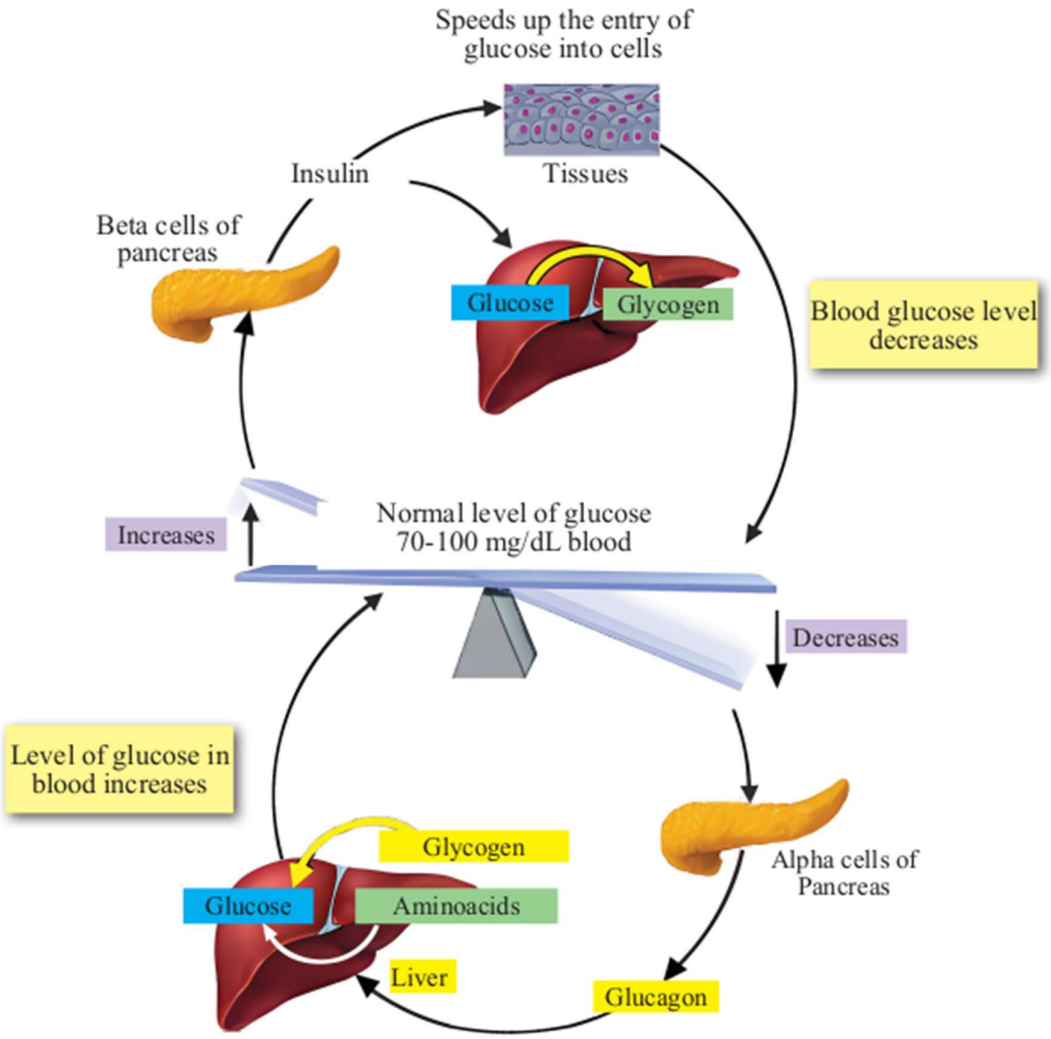


Illustration 4.7 Regulation of glucose in blood

ADRENAL GLAND

Position – Seen as a cap above kidney

Parts - Medulla (Internal Part)

Cortex (External part)

HORMONES SECRETED BY MEDULLA

- Epinephrine
- Nor epinephrine

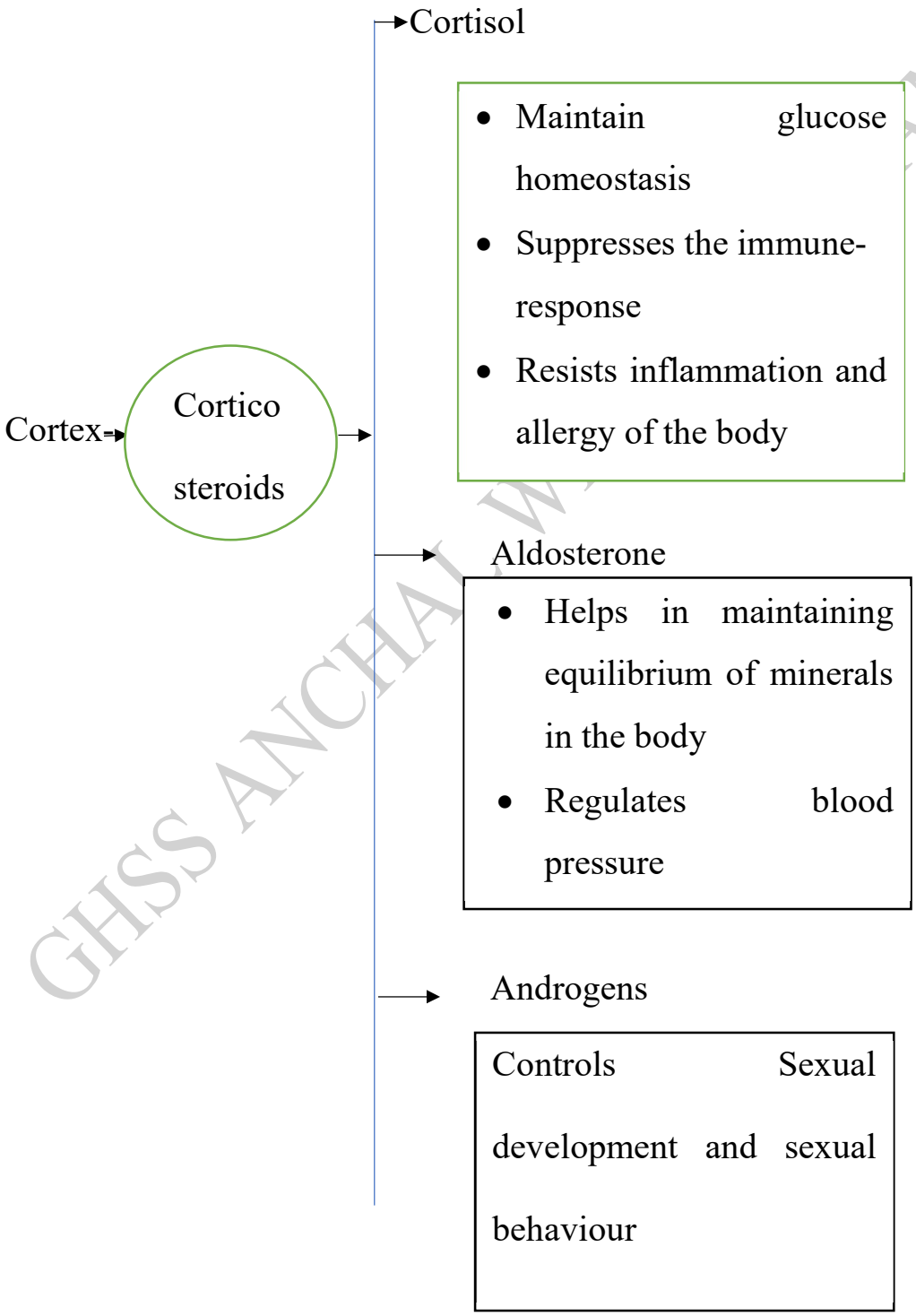
HORMONES SECRETED BY CORTEX

- Glucocorticoid
- Mineralo corticoid
- Gonado Corticoid

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They are generally called Corticosteroids

- **HORMONES PRODUCED BY CORTEX AND THEIR FUNCTIONS.**



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THE WAY TO OVERCOME. EMERGENCY BY EPINEPHRINE, NOREPINEPHRINE

- Increases heart Rate and blood pressure
- Increased blood glucose level
- Increased blood flow to the heart and muscles
- Decreased Blood flow to skin and digestive organ these hormones

act with Sympathetic Nervous system in Emergency Situations

PINEAL GLAND

Position - located near the thalamus in the human brain

Hormone - Melatonin.

Function - Melatonin plays a major role in controlling the activities that occur at regular intervals throughout the day.

The pineal gland is known as biological clock

THYMUS GLAND

Position - Seen Behind Sternum

Hormone - Thymosin

Helps in Maturation of T lymphocytes

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HYPOTHALAMUS AND PITUITARY - MASTER CONTROL SYSTEM

Hypothalamus.

- The functions of most endocrine glands are Controlled by Hypothalamus
- Hypothalamus maintain homeostasis
- Hypothalamus is the master controller of hormone production process.

Cells - Neuro Secretory Cells

- **Hormones**
    - Vasopressin
    - Oxytocin
    - Releasing hormones
    - Inhibitory hormones

FUNCTIONS OF HORMONES SECRETED BY HYPOTHALAMUS

- **Vasopressin (ADH)-** Reabsorption of water from kidney

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- **Oxytocin** -Lactation (production of Milk)
  - Contraction of smooth muscle
- **Releasing hormone :**

  Stimulates the production of trophic hormones.
- **Inhibitory hormone**

  – Inhibit the production of trophic hormones

PITUITARY GLAND

Position : Seen behind Hypothalamus.

Parts :

- Anterior lobe
- Posterior lobe

HORMONES SECRETED BY ANTERIOR LOBE OF PITUITARY

TROPHIC HORMONES.

- TSH
- ACTH
- GTH
- Somatotrophin tropin
- prolactin

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- TSH - Stimulates thyroid gland
- ACTH -Stimulates Adrenal cortex
- GTE -Stimulates the action of ovary and testes
- Somatotrophin - Helps in Body growth
- prolactin - produces milk.

POSTERIOR LOBE OF PITUITARY

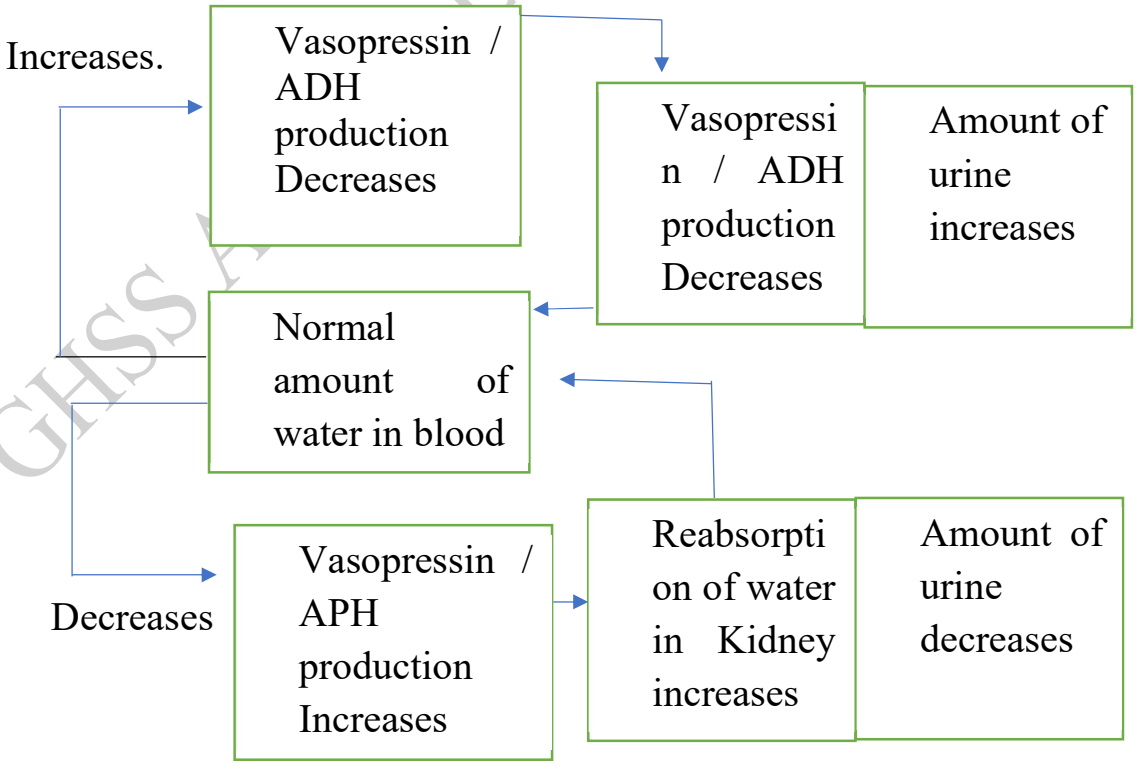
- Stores the hormones vasopressin and Oxytocin Secreted by hypothalamus
- Secretes oxytocin and vasopressin When required

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IMPORTANT GROWTH DISORDERS

Disorder	Cause	Symptoms
Dwarfism	Production of Somatotropin decreases during the growth phase	Stunted growth
Gigantism	The production of Somatotropin Increases during growth phase	Excessive body growth
Acromegaly	The production of Somatotropin increases after growth phase	Excessive body growth body parts such as palms foot and jaws

WATER BALANCE(ADH)



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REPRODUCTIVE HORMONES AND THEIR FUNCTIONS

Hormone	Gland	Function
Testosterone	Testes	• Activity of male sex organs • Sperm production • Control secondary sexual characteristics
Estrogen	Ovary	• Activity of female sex organs • Production of ovum • Menstrual cycle • Control Secondary sexual characteristics
Progesterone	Ovary	• Maintain the foets in the uterus • Menstrual cycle
Prolactin	Anterior lobe of Pituitary	Produces milk
Oxytocin	Hypothalamus	• Lactation • contraction of smooth Muscle

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NEUROTRANSMITTERS AET AS HORMONES

- Dopamine
- Serotonin.
- endorphins.

PHEROMONES

Pheromones are Chemical substances that are Secreted by organisms to their immediate surroundings to facilitate Communication with the Same Species.

DIFFERENT TYPE OF PHEROMONES AND THEIR FUNCTIONS

Different type of Pheromones	Function
Territorial pheromone	To determine territory
Aggregation pheromone	To ensure availability of food
Alarm pheromone	Signaling dangers.
Trail pheromones	For trail marking
Sex pheromones	To attract mates

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CHAPTER 5

IMMUNITY AND HEALTHCARE

Antigens : The substances that cause disease

eg: Pathogens, pollengrains, dust, chemicals, certain food items etc.

Immunity : The ability of the body to prevent the entry of pathogens.

Immunity are of 2 types.

(1) Innate immunity 2) Acquired Immunity

I. INNATE IMMUNITY [NON-SPECIFIC IMMUNITY]

a. Primary level Immunity.

- Natural immune mechanism exist in the body by birth.

Body Part	**Immune Mechanism**
Mouth	Lysozyme enzyme (Saliva)
Eye	Lysozyme enzyme (tear
Trachea	Cilia
Ear	Wax
Stomach	Hcl
Skin	Sweat and sebum
Digestive tract and urinary tract	Useful bacteria

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b. Secondary level immunity

- If the pathogens break the primary level immunity and enter the body, the immune system against them is called Acquired immunity.
- White blood cells are involved

1. Neutrophil - Engulf and destroys the bacteria.

- Produce chemicals against them.

2. Monocyte - Engulf and destroys the bacteria.

- Stimulate other immune cells.

3. Eosinophil - Neutralizes the chemicals released due to allergy.

- destroy parasites.

4. Basophil - Produces heparin and histamine.

- heparin → Prevent blood clotting.

- histamin → cause inflammation.

ACTION OF HISTAMIN

- Histamin dilates the blood vessel.
- So blood flow increases, WBC reach the injured area.
- They engulf and destroy the pathogens.

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PHAGOCYTOSIS

- Process by which WBC engulf and destroy pathogens.

  eg: neutrophils, monocytes.

FEVER

- A kind of immune mechanism.
- When the body temperature rises above the normal range (98.6^0 f / 37^0 c) is known as fever.

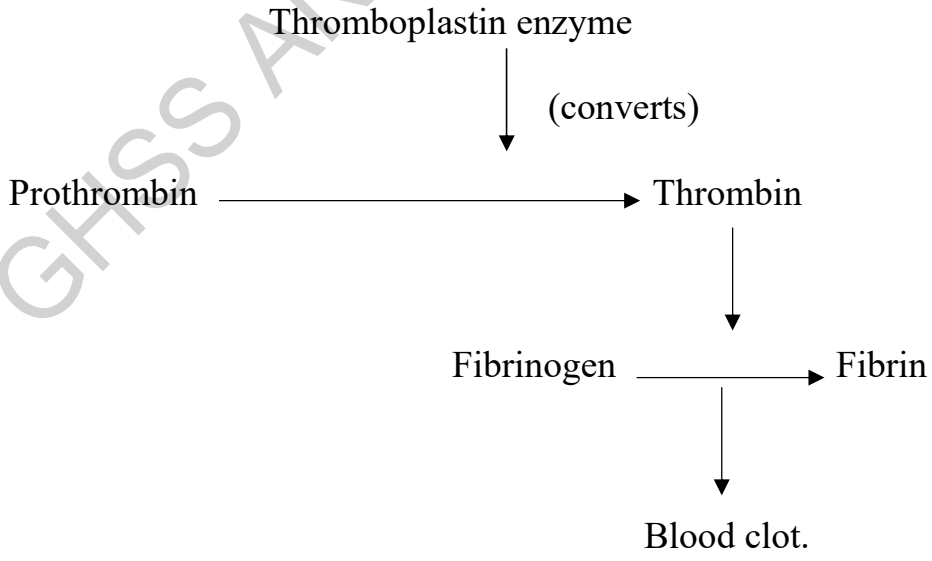
PYROGENS

- When infection occurs ,WBC release chemical substances called pyrogens.
- Pyrogens stimulate the brain to raise the body temperature.
- High temperature helps the phagocytes to destroy pathogens more quickly.

BLOOD CLOTTING

- Mechanism to prevent the flow of blood and block the entry of pathogen.

MECHANISM



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- When an injury occurs, platelets and damaged tissue produce an enzyme called **Thromboplastin.**
- A clot forms a network of fibres that trap blood cells, forming blood clot.

II .ACQUIRED IMMUNITY [SPECIFIC IMMUNITY]

- Immunity that develops after birth.
- Specifically identifies and defends antigens that enter the body.
- Carried out by **T lymphocytes** and **B lymphocytes.**

Characteristics	T Lymphocytes	B Lymphocytes
Formation	bone marrow.	bone marrow.
Maturation	bone marrow.	Thymus gland.
Function	Produce **antibodies** act against pathogens	Identify and destoy the infected cells and **cancer cells**

- **Father of vaccination — Edward Jenner**

ARTIFICIAL IMMUNITY

Immunity acquired by immunisation.

IMMUNISATION

Artificial method of preparing the immune cells in advance, expecting the attack of pathogens through vaccination.

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Vaccine	Components of vaccine	Preventable disease
MMR	neutralised pathogen	Measles, mumps, Rubella
OPV (Oral Polio vaccine)	dead pathogens	Poliomyelitis (Polio)
HPV	Parts of pathogen	cervical cancer
Covid-19 vaccine	parts of the genetic material of the pathogen	Covid 19

BACTERIAL DISEASES

Characteristics	**Tuberculosis**	**Leptospirosis (Rat fever)**
Pathogen	**Mycobacterium tuberculosis**	**Leptospira**
Mode of transmission	Coughing, sneezing, talking release tiny droplets into the air	Through wounds, through nose from water contaminated with rat urine
Part of the body affected	Lung, kidney, bone, brain	Kidney, liver, lung

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Symptoms	Persistent cough, fever, loss of appetite, weight loss	Sudden fever, headache, muscle pain, redness of eye, vomiting
Prevention	BCG vaccination, DOTS	Antibiotics

VIRAL DISEASES

1.AIDS (ACQUIRED IMMUNO DEFICIENCY SYNDROME)

Pathogen → HIV (Human Immunodeficiency Virus)

HIV affects → T lymphocytes

Symptoms → Immunity decreases, pneumonia, tuberculosis, cancer

Mode of transmission →

→ Sexual contact with an infected person
→ Sharing needles and syringes
→ Transfusion of blood or organs containing HIV
→ From infected mother to foetus.

Treatment → Antiretroviral Therapy (ART)

2. NIPAH

Pathogen → Nipah virus

Symptoms → Fever, headache, cough, vomiting, epilepsy, loss of consciousness

Mode of transmission→ From infected animals (bat, pig) to humans through Contaminated food.

Vaccine → No vaccine, only supportive care

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FUNGAL DISEASE

CHARACTERISTICS	RINGWORM	CANDIDIASIS
Mode of infection	• Through infected person • Contaminated objects	• Sharing of clothes • Direct contact • Through sexual contact
Symptoms	• Circular reddish rashes • Itching	• White patches in the mouth • Itching

PROTOZOANS

They are unicellular eukaryotes.

1. MALARIA

Pathogen → Plasmodium

Transmitted by → Female Anopheles mosquito

Host → Humans

Affect → Destroys the red blood cells

Symptoms → Intermittent fever, chills, shivering, body pain, muscle pain

Treatment → Antimalarial medicine

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2. AMOEBIC ENCEPHALITIS

Pathogen → Naegleria fowleri (Amoeba)

Mode of transmission → Contaminated water enters the nose and reaches the brain

Preventive measure → Avoiding bathing in or washing the face with contaminated water

WORMS

1. WORM INFESTATIONS

Pathogens → Pinworms, roundworms

Transmitted by → Nail biting,eating without washing hands,Drinking contaminated water

Symptoms → Dizziness, vomiting, weight loss, fatigue

Treatment → Albendazole tablet

2. FILARIASIS

Pathogen → Filarial worms

Affect → Lymphatic vessels

Transmitted by → Female Culex mosquitoes

Symptoms → Legs, Arm, Genital Organs become swollen

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CANCER

- Abnormal and uncontrolled cell division leads to multiplication of cells causing cancer.
- Cancer cells spread through blood and lymph.

Causes →Genetic change, Lifestyle, Viral infection, Environmental factors

Treatment →Surgery, Chemotherapy, Radiation therapy, Hormone therapy

Type	Disease
Life style disease	Diabetes mellitus, Hypertension
Occupational disease	Silicosis
Autoimmune disorders	Rheumatoid arthritis
Nutritional disorder	Anaemia

HEREDITARY DISEASES

Caused by genetic defects

Sickle Cell Anaemia	Haemophilia
• Defect of the gene that controls the production of haemoglobin • **Red blood cells become sickle-shaped** • Oxygen carrying capacity decreases • They get stuck in blood vessels and block blood flow	• Defect of the gene that controls the production of proteins for blood clotting • Even small wounds cause excessive bleeding

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Treatment

Ayurveda → Originated in India

Father of Ayurveda → **Charaka**

Treatment → Diet regulation, exercise, herbal medicines

Homeopathy - Founded by **Samuel Hahnemann**

Treatment → Use diluted doses of natural medicines

Modern Medicine → **Hippocrates**

Gives importance to diagnosis and medicine

ANTIBIOTICS

Medicines derived from microorganisms or produced synthetically, used against bacteria.

SIDE EFFECTS

- ❖ Digestive problem
- ❖ Allergy problem
- ❖ Damages to liver, kidney
- ❖ Resistance against antibiotics.

First Antibiotic - Penicillin from Penicillium notatum (fungi)

Discovered by - Alexander Fleming

BLOOD DONATION

- Age between 18 and 60 years
- Minimum weight 50 kg

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- Hemoglobin level should be above 12.5 g/dL
- Donate blood once every 3 months for a healthy man and once every 4 months for a healthy woman
- Do not donate blood during menstruation and breast-feeding time

BLOOD TRANSFUSION

- The process of transferring blood from one person to another
- Who gives blood → Donor
- Who receives blood → Recipient
- Blood groups are determined by the presence of A or B antigens on the surface of RBC
- D antigen present → Blood group is Rh positive (+ve)
- D antigen absent → Blood group is Rh negative (–ve)
- D antigen is known as **Rhesus factor**
- D antigen was discovered in Rhesus Monkeys

BLOOD GROUP TABLE

Blood Group	Antigen	Antibody
A	A	b
B	B	a
AB	A, B	–
O	–	a, b

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DIAGNOSIS

Equipment	Use
Stethoscope	Detecting heart and lung sounds
Sphygmomanometer	To check blood pressure
Digital BP apparatus	To check blood pressure
Pulse oximeter	To check blood oxygen saturation
Thermometer	Measure temperature
Endoscope	To take pictures or videos of internal organs
US (Ultrasound) scanner	To take real pictures of internal organs
CT scanner	To check bone fracture, internal bleeding, tumours, etc.
MRI scanner	Detailed image of soft tissue structure

Cardiology → Department of Heart

Paediatrics → Department of children

Gynaecology → Female reproductive system

Nephrology → Kidney

Neurology → Brain

ENT → Ear, Nose, Throat

Urology → Urinary tract, male reproductive system

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ANIMAL DISEASE

Bacterial Disease	Viral Disease
• Anthrax – cattle • Mastitis – cattle	• Foot and mouth disease → cattle • Ranikhet Disease → chicken, duck

Plant Disease

Virus	Bacteria	Fungi
• Bunchy top of banana • Mosaic disease in Tapioca	• Wilt of brinjal • Blight disease of paddy	• Bud rot of coconut • Quick wilt of pepper

IMMUNITY IN PLANTS

- Rigid cell wall
- Presence of epidermis
- Waxy coating on leaves
- Presence of antimicrobial substance
- Plant enzymes breakdown the pathogen
- A poly saccharide called callose prevent the spread of pathogen.

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CHAPTER 6

BIOLOGY AND TECHNOLOGY

BIOTECHNOLOGY

- Technology of using living things to provide products useful to humans.

GENETIC ENGINEERING

- Technologies for creating desired traits in organisms, by combining or deletin genes

RECOMBINANT DNA TECHNOLOGIES

- Combines DNA segments from two or more different organisms, to form recombinant DNA.

Genetic Scissor - **Restriction Endonuclease enzyme**

- Cuts a specific gene.

Genetic Glue - **Ligase enzyme**

- joins two DNA segments

vector -**Plasmids** (circular DNA in Bacteria) and **virus**

- That carries the Recombinant DNA to the host

Host cell - The organism where recombinant DNA is inserted.

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CRISPER TECHNOLOGY

- Used to edit DNA
- Guide RNA (gRNA) – RNA that identifies the DNA segment to be cut.
- Cas 9 enzyme - Cut the DNA

STEPS

Identification - Guide RNA identifies the DNA to be cut

Cutting - Cas 9 enzyme cuts the DNA

Editing - The nitrogen base sequences are edited

GMO - GENETICALLY MODIFIED ORGANISMS.

Organisms whose genetic structure is altered through genetic engineering

Group of organism	Characteristics	Examples
Plants	Pest resistance	Bt cotton
	Nutrient enriched	Glodenrice
	Weedicide resistance	Soyabean
	Virus resistance	RSVresistant pappaya
Animals	Fast growth , high productivity, disease resistance	Salmon fish
	Fluorescence	Glo fish

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Micro organism	Decomposition ability	Psuedomonas putida
	Produce insulin	E coli
	Farmentation ability	Baker's yeast
	High Nitrogen fixing ability	Rhizobium

Genome

Sum total of all the genetic materials in an organism

HUMAN GENOME PROJECT (HGP)

- International research project
- Period - 1990- 2003
- Goal – Understanding the information in the human genome.

RESULTS

- 300 crore base pairs in human genome
- 20,000 – 25,000 genes in humans
- Functional genes → 1–2%
- Non-functional genes →98%, called junk DNA

GENE MAPPING TECHNOLOGY

- Identifies the location of genes in DNA
- Used in Human Genome Project

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GENE THERAPY

- Method of treating genes
- Removal of defective genes and replacing them with functional genes

Example: Sickle cell Anaemia

DNA FINGERPRINTING

- Discovered by: Alec Jeffrey
- Used to Analyse the sequence of nucleotides in DNA and identify individuals accordingly is called DNA fingerprinting
- The variations occur in the non-coding regions of DNA

THE BASIS OF DNA FINGERPRINTING

- The variations occur in the non-coding regions of DNA. The number of repeats and their order differ in each individual
- Each individual has a unique DNA pattern, as unique as the fingerprint
- The DNA patterns of parents and offspring are more similar

USES OF DNA FINGERPRINTING

- To identify culprits
- To identify genetic disorders
- To determine the relationship between parents and offspring
- To identify victims of accidents or disasters

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HUMAN MICROBIOME

The collection of microorganisms and their genes that inhabit the human body

HUMAN MICROBIOME PROJECT (HMP)

- Launched by the National Institute of Health in 2007
- The project proved that the human body is not only composed of cells but also crores microorganisms
- It revealed that the cause of disease is not only pathogens, but also changes in the equilibrium of microorganisms in the body
- Helps to develop personalised medicine

APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN BIOTECHNOLOGY

1. Diagnosis and treatment of diseases
2. Development of personalised medicine
3. Genome sequencing
4. Improving agriculture
5. Predicting air and water pollution levels

6. **BIOINFORMATICS**

- New branch of science
- Analyses the biological data by combining computer science, mathematics, and science

Technological Applications in Environmental Conservation

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1. Wildlife tracking sensors

2. Global biodiversity data sensors

3. **Bioremediation**

- Removal of pollutants from the environment by using microorganisms and plants
- Converts pollutants into harmless compounds

4. **Cryopreservation**

Process of preserving living cells, tissues, and organs at very low temperature (−196°C)

THE GREAT PACIFIC GARBAGE

- In the Pacific Ocean, plastic has accumulated and formed a floating region of 16 lakh square km

❖ **Environment Day: June 5**

❖ **World Ocean Day: June 8**

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